

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalogafica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its

`\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one

`foreignabstract (env.)` page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for

`\printlosymbols`

`\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains $\text{\LaTeX}$ commands, mathematical symbols, or mix of uppercase and lowercase. `MakeIndex` needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain `MakeIndex` input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

`COPPE $\text{\TeX}$` produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no `BIB $\text{\TeX}$` `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` `BIB $\text{\TeX}$` styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `L $\text{\TeX}$` , `biber`, and twice again `L $\text{\TeX}$` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephD, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Latin Modern* (`\lmodern`), which is what the class loads. Redefine if you loaded `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level **a-2b** and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level a-2b rather than a-1b: a-1b forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

assinaturas Give every member of the board a signature rule of its own on the folha de aprovacao, as Annex D of the manual shows, instead of the compact list. The spacing adapts to the size of the board.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit:** the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites T_EX has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdfT_EX the class therefore loads the `morewrites` package by itself when it is installed, which removes the ceiling. **semmorewrites** turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. **morewrites** means “I insist”: a missing package becomes an error instead of a warning. Under LuaLaT_EX, which has 128 streams, none of this is loaded and none of it is needed.

semserifa Set the whole document in the sans-serif family instead of the serif one. **Neither the manual nor the COPPE norm prescribes a typeface:** 2.2(b) fixes the colour, the body size 12 and the smaller uniform size of the four items it lists, and says nothing about the family. The choice is the author’s, and the SiBI manual that carries the rule is itself set in Arial. The

class loads `lmodern`, which brings both families as vector fonts, so either way the document is PDF/A material. For a DIFFERENT sans face – Helvetica, say – load its package in the preamble and redefine `\familydefault` there; this option is for the common case of an author who does not want to choose a font, only to drop the serifs. Redefine `\coppefinalfont` if you do load another face, so the colophon does not announce Latin Modern.

listasnosumario Put the pre-textual lists – figures, tables, quadros, programas, abreviaturas, simbolos – back into the sumario. **Contrary to 3.1.2.1.6**, which tells the author to follow the manual’s own sumario, and that one opens at “1 INTRODUCAO”: the lists are pre-textual elements and come BEFORE the sumario, so listing them inside it is listing what the reader has already passed. The option exists because every CoppeTeX document written before this release had them there, and a work in its final weeks should not have its sumario change under it.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals. **Contrary to 2.7**, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (`brazilian` by default; one of `english`, `spanish`, `french`, `italian` also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein` Two pieces of user-facing API come with the multilingual model:
`brazilianabstract (env.)`
`\usecoppelanguage`

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesis`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T\'i tulo em portugu~es} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T\'i tulo en espa~nol} % cover/folha
  \author{Juan}{P\'erez}
  \advisor{Prof.}{Mar\'i a}{Garc\'i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilinguaje}
  \maketitle
  \frontmatter
  \begin{abstract} Resumen principal en espa~nol. \end{abstract}
  \begin{foreignabstract} English foreign abstract. \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\'on} Cuerpo en espa~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your TEXINPUTS path (e.g. inside the local texmf tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom biblatex datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` biblatex bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` biblatex localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
```

`\DeclareLanguageMapping{english}{english-coppe}`

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are

already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between `pdfTeX` and the Unicode engines (`LuaTeX`/`XeTeX`).

`inputenc (utf8)` UTF-8 input; loaded *only under pdfTeX* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfTeX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms; the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of `biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles (`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is `biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}`, ...).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_latex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfLaTeX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating} % rodando coisas, como tabelas
```



```

5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz} % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23 % Fonte: a norma fixa o corpo 12, e não a família. Para compor sem serifa,
24 % passe a opção de classe 'semserifa' lá em cima -- NÃO redefina
25 % \familydefault aqui, porque isso venceria a opção. Para uma sem serifa
26 % diferente da Latin Modern, carregue o pacote da fonte e troque a família:
27 %\usepackage{helvet}\renewcommand{\familydefault}{\sfdefault}
28
29
30 \begin{document}
31 \title{Título da Tese}
32 \foreigntitle{Thesis Title}
33 \author{Nome do Autor}{Sobrenome}
34 \advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
35 \advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
36 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
37 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
38 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
39 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
40
41 \examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
42 \examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
43 \examiner[UFF]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
44 \examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
45 \examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
46 \department{PESC}
47 \date{06}{2026}
48
49 \keyword{Primeira palavra-chave}
50 \keyword{Segunda palavra-chave}
51 \keyword{Terceira palavra-chave}
52
53 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
54 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
55 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
56 \linhapesquisa{Engenharia de Dados e Conhecimento}
57 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
58 \projetovinculado{sim} % sim | nao

```

```

59 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
60 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
61 Tecnologico}{CNPq}
62 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
63 Estado do Rio de Janeiro}{FAPERJ}
64 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
65 % descomente a linha abaixo e a moldura vazia da folha adicional some.
66 % \fichacatalografica{ficha.pdf}
67
68 \maketitle
69
70 \frontmatter
71 \dedication{A alguém cujo valor é digno desta dedicatória.}
72
73 \chapter*{Agradecimentos}
74
75 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
76 orientadores, colegas, familiares e \as ag\encias de fomento. Lorem ipsum
77 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
78 ut labore et dolore magna aliqua.
79
80 \begin{abstract}
81
82 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
83 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
84 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
85 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
86 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
87 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
88 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
89 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
90 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
91 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
92 aspernatur aut odit aut fugit.
93
94 \end{abstract}
95
96 \begin{foreignabstract}
97
98 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
99 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
100 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
101 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
102 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
103 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
104 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
105 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
106 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
107 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
108 aspernatur aut odit aut fugit.
109
110 \end{foreignabstract}
111
112 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,

```

```

113 % abreviaturas e siglas, símbolos) ANTES do sumario; sumario por ultimo.
114 \listoffigures
115 \listoftables
116 \listofquadros
117 \listofprogramas
118 \listofalgorithms
119 \printloabbreviations
120 \printlosymbols
121 \tableofcontents
122
123 \mainmatter
124 \chapter{Introdução}
125
126 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos do
127
128 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
129 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
130
131 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
132
133 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
134
135 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
136
137 Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/CO
138
139 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d
140
141 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
142
143 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
144
145 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
146
147 Este documento não substitui, mas complementa, o documento que descreve a classe.
148
149
150
151 \chapter{Configurações Iniciais}
152
153 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
154
155 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
156
157 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
158
159 \section{Linguagem principal do texto}
160
161 Essa classe considera que o texto principal está em português e algumas partes específicas
162 \begin{itemize}
163 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
164 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
165 \end{itemize}
166

```

167 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CoppET
168
169
170 \section{Por que usar o \LaTeX}
171
172 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
173
174 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
175
176 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
177
178 \section{Como e onde usar o \LaTeX}
179
180 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
181 \begin{enumerate}
182 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
183 \item Usar um ambiente na rede, como o Overleaf.
184 \end{enumerate}
185
186 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
187
188
189 \chapter{Algumas Regras da COPPE}
190
191 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
192
193 É imprescindível definir os símbolos, tal como o
194 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
195 \syml{\mathbb{R}}{Conjunto dos números reais}
196 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
197
198 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
199
200 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
201
202 \section{Citações}
203 \label{sec:citacoes}
204
205 Citações curtas podem ser feitas \enquote{o comando quote} ou direto com “duas crases e do
206
207 \begin{longquote}
208 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
209 feita com o ambiente \verb*|longquote| The primary objective of this
210 investigation was to determine the feasibility of detecting corrosion in
211 aluminum Naval aircraft components with neutron radiographic interrogation
212 and the use of standard corrosion penetrameters. Secondary objectives
213 included the determination of the effect of object thickness on image quality,
214 the defining of minimum levels of detectability and a preliminary investigation
215 of a means whereby the degree of corrosion could be quantified with neutron
216 radiographic data. \citep{article-example}
217 \end{longquote}
218
219 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
220

221 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em

222

223 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref

224

225 \chapter{Floats}

226

227 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \textit

228

229 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm

230

231 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração

232

233 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao

234

235

236 \section{Tabelas e Figuras Padrão}

237

238 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.

239

240 \begin{table}[ht]

241 \centering % Centraliza a tabela

242 \caption{Exemplo de Tabela de Números}

243 \label{tab:exemplo_numeros}

244 \begin{tabular}{ccc} % Define a quantidade de colunas

245 \hline % Linha superior

246 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos

247 \hline % Linha média

248 1 & 2 & 3 \\ % Primeira linha de dados

249 \hline

250 4 & 5 & 6 \\ % Segunda linha de dados

251 \hline

252 7 & 8 & 9 \\ % Terceira linha de dados

253 \hline

254 10 & 11 & 12 \\ % Quarta linha de dados

255 \hline % Linha inferior

256 \end{tabular}

257 \fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float

258 \end{table}

259

260

261

262 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.

263

264 \begin{figure}[ht]

265 \centering % Centraliza a figura

266 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)

267 \label{fig:exemplo_figura} % Etiqueta para referência cruzada

268 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura

269 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura

270 \end{figure}

271

272 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.

273

274 \begin{quadro}[ht]

```

275 \centering
276 \caption{Exemplo de Quadro}
277 \label{qua:exemplo}
278 \begin{tabular}{|l|l|}
279 \hline
280 \textbf{Ilustração} & \textbf{Delimitação} \\
281 \hline
282 Tabela & Linhas apenas no topo e na base. \\
283 \hline
284 Quadro & Linhas em todas as bordas. \\
285 \hline
286 \end{tabular}
287 \fonte{Elabora\c{c}\~ao pr\'opria.}
288 \end{quadro}
289
290
291
292
293
294 \section{Tabelas mais elegantes}
295
296 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
297
298 \begin{table}[ht]
299 \centering % Centraliza a tabela
300 \caption{Exemplo de Tabela de Números mais elegantes}
301 \label{tab:exemplo_numerosbom}
302 \begin{tabular}{ccc} % Define a quantidade de colunas
303 \toprule % Linha superior
304 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\
305 \midrule % Linha média
306 1 & 2 & 3 \\
307 4 & 5 & 6 \\
308 7 & 8 & 9 \\
309 10 & 11 & 12 \\
310 \bottomrule % Linha inferior
311 \end{tabular}
312 \fonte{Elabora\c{c}\~ao pr\'opria.}
313 \end{table}
314
315 \section{Tabelas Longas ou Largas}
316
317 Se sua tabela é muito longa ou larga, existem várias opções.
318 \begin{itemize}
319 \item alterar o tamanho da letra
320 \item Usar o longtable
321 \item rodar a tabela, fazendo ela em \textit{landscape}
322 \item fazer a tabela dentro de um minibox
323 \end{itemize}
324
325
326 \subsection{Tabelas largas demais}
327
328 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.

```

```

329
330 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
331
332 \begin{table}[ht]
333 \centering % Centraliza a tabela
334 \caption{Exemplo de Tabela Larga com Fonte Menor}
335 \label{tab:tabela_largafns}
336 \footnotesize % Aplica uma fonte menor para a tabela
337 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
338 \toprule
339 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
340 \midrule
341 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
342 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
343 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
344 \bottomrule
345 \end{tabular}
346 \fonte{Elabora\c{c}\~ao pr\'opria.}
347 \end{table}
348
349 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
350
351 \begin{table}[ht]
352 \centering
353 \caption{Exemplo de Tabela Redimensionada}
354 \label{tab:examplerb}
355 \resizebox{\textwidth}{!}{%
356 \begin{tabular}{llll}
357 \toprule
358 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
359 \midrule
360 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
361 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
362 \bottomrule
363 \end{tabular}%
364 }
365 \fonte{Elabora\c{c}\~ao pr\'opria.}
366 \end{table}
367
368
369 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat
370
371 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
372
373 \begin{sidewaystable}
374 \centering
375 \caption{Sua Legenda Aqui}
376 \label{tab:sua_tabela}
377 \begin{tabular}{lll}
378 \toprule
379 Coluna 1 & Coluna 2 & Coluna 3 \\
380 \midrule
381 Item 1 & Item 2 & Item 3 \\
382 Item 4 & Item 5 & Item 6

```

```

383 \bottomrule
384 \end{tabular}
385 \fonte{Elabora\c{c}\~ao pr\'opria.}
386 \end{sidewaystable}
387
388 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
389
390 % Exemplo de tabela longa que se estende por várias páginas
391 \begin{longtable}{|c|c|c|}
392 % primeiro cabeçalho (é o caption)
393 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
394 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
395 \endfirsthead
396 % cabeçalho normal
397 \multicolumn{3}{c}%
398 {\table\ \thetable} -- continuação da página anterior} \\
399 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
400 \endhead
401 % pé normal
402 \hline \multicolumn{3}{r|}{Continua na próxima página} \\ \hline
403 \endfoot
404 \hline
405 % último pé
406 \multicolumn{3}{r|}{Continua na próxima página}} \\
407 \hline \hline
408 \endlastfoot
409
410 % Conteúdo da tabela
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
418 4 & 5 & 6 \\
419 1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3 \\
422 4 & 5 & 6 \\
423 1 & 2 & 3 \\
424 4 & 5 & 6 \\
425 1 & 2 & 3 \\
426 4 & 5 & 6 \\
427 1 & 2 & 3 \\
428 1 & 2 & 3 \\
429 4 & 5 & 6 \\
430 1 & 2 & 3 \\
431 4 & 5 & 6 \\
432 1 & 2 & 3 \\
433 4 & 5 & 6 \\
434 1 & 2 & 3 \\
435 4 & 5 & 6 \\
436 1 & 2 & 3

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437 4 & 5 & 6 \\
 438 1 & 2 & 3 \\
 439 4 & 5 & 6 \\
 440 1 & 2 & 3 \\
 441 4 & 5 & 6 \\
 442 1 & 2 & 3 \\
 443 4 & 5 & 6 \\
 444 1 & 2 & 3 \\
 445 4 & 5 & 6 \\
 446 1 & 2 & 3 \\
 447 4 & 5 & 6 \\
 448 1 & 2 & 3 \\
 449 4 & 5 & 6 \\
 450 1 & 2 & 3 \\
 451 4 & 5 & 6 \\\ 1 & 2 & 3 \\
 452 4 & 5 & 6 \\
 453 1 & 2 & 3 \\
 454 4 & 5 & 6 \\
 455 1 & 2 & 3 \\
 456 1 & 2 & 3 \\
 457 4 & 5 & 6 \\
 458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
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 485 4 & 5 & 6 \\
 486 1 & 2 & 3 \\
 487 4 & 5 & 6 \\
 488 1 & 2 & 3 \\
 489 4 & 5 & 6 \\
 490 1 & 2 & 3 \

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491 4 & 5 & 6 \\
492 1 & 2 & 3 \\
493 4 & 5 & 6 \\
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508 1 & 2 & 3 \\
509 4 & 5 & 6 \\
510 1 & 2 & 3 \\
511 4 & 5 & 6 \\
512 1 & 2 & 3 \\
513 4 & 5 & 6 \\
514 1 & 2 & 3 \\
515 4 & 5 & 6 \\
516 1 & 2 & 3 \\
517 4 & 5 & 6 \\
518
519 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
520 \end{longtable}
521 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
522
523 \section{Subfiguras (subcaption)}
524
525 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
526
527 \begin{figure}[ht]
528   \centering
529   \begin{subfigure}[b]{0.42\textwidth}
530     \centering\includegraphics[width=\linewidth]{example-image-a}
531     \caption{Primeira vista}\label{fig:sub-a}
532   \end{subfigure}
533   \hfill
534   \begin{subfigure}[b]{0.42\textwidth}
535     \centering\includegraphics[width=\linewidth]{example-image-b}
536     \caption{Segunda vista}\label{fig:sub-b}
537   \end{subfigure}
538   \caption{Figura com duas subfiguras}
539   \label{fig:subs}
540   \fonte{Elabora\c{c}\~ao pr\'opria.}
541 \end{figure}
542
543 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
544 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.

```

```

545
546 \section{Imagens (graphicx)}
547
548 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
549 \begin{itemize}
550   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
551   \item \verb|[angle=90]| --- rotação;
552   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, dire
553 \end{itemize}
554 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt
555
556 \section{Introdução ao tikz}
557
558 O \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
559
560 \begin{figure}[ht]
561   \centering
562   \begin{tikzpicture}
563     \node[draw, rounded corners] (a) at (0,0) {Início};
564     \node[draw, rounded corners] (b) at (4,0) {Fim};
565     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
566   \end{tikzpicture}
567   \caption{Diagrama simples em tikz}
568   \label{fig:tikz}
569   \fonte{Elabora\c{c}\~ao pr'opria.}
570 \end{figure}
571
572 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
573
574 \section{Float ou direto no texto?}
575
576 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|L
577
578 \chapter{Using BibLaTeX}
579 \label{cap:biblatex}
580
581 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
582
583 \textbf{É} obrigatório compilar com \texttt{biber}} (e não com \hologo{BibTeX}). A sequênci
584
585 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
586
587 \begin{table}[h]
588 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
589   \texttt{\textbackslash cite} do \LaTeX\ e
590   o comando \texttt{\textbackslash citep},
591   disponível pela opção \texttt{natbib} do BibLaTeX.}
592 \label{tab:citation}
593 \centering
594 {\footnotesize
595 \begin{tabular}{|c|c|c|}
596   \hline
597   Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
598   \hline

```

```

599 Livro & \cite{book-example} & \citep{book-example}\\
600 Artigo & \cite{article-example} & \citep{article-example}\\
601 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
602 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
603 Anais de Congresso & \cite{inproceedings-example} &
604 \citep{inproceedings-example}\\
605 Séries & \cite{incollection-example} & \citep{incollection-example}\\
606 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
607 Dissertação de mestrado & \cite{mastersthesis-example} &
608 \citep{mastersthesis-example}\\
609 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
610 \hline
611 \end{tabular}}
612 \fonte{Elabora\c{c}\~ao pr\'opria.}
613 \end{table}
614
615 \begin{table}[h]
616 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
617 \texttt{\textbackslash cite} do \LaTeX\ e
618 o comando \texttt{\textbackslash citet},
619 disponível pela opção \texttt{\natbib} do BibLaTeX. Além disso, usando o booktabs.}
620 \label{tab:citation1}
621 \centering
622 {\footnotesize
623 \begin{tabular}{ccc}
624 \toprule
625 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
626 \midrule
627 Livro & \cite{book-example} & \citet{book-example}\\
628 Artigo & \cite{article-example} & \citet{article-example}\\
629 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
630 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
631 Anais de Congresso & \cite{inproceedings-example} &
632 \citet{inproceedings-example}\\
633 Séries & \cite{incollection-example} & \citet{incollection-example}\\
634 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
635 Dissertação de mestrado & \cite{mastersthesis-example} &
636 \citet{mastersthesis-example}\\
637 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
638 \bottomrule
639 \end{tabular}}
640 \fonte{Elabora\c{c}\~ao pr\'opria.}
641 \end{table}
642
643 \section{Comandos de citação}
644
645 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
646
647 \subsection{Parâmetro opcional: página e “tradução nossa”}
648
649 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
650
651 \subsection{Citação de citação (\emph{apud})}
652

```

```

653 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
654
655 \subsection{Múltiplas bibliografias}
656
657 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
658
659 \section{Tipos de entrada bibliográfica}
660
661 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
662
663 \subsection{Tipos acadêmicos principais}
664 \begin{itemize}
665 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
666 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \v
667 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
668 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
669 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
670 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); períod
671 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
672 \end{itemize}
673
674 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
675 \begin{itemize}
676 \item Patente --- \verb|@patent| (\verb|@patente|)
677 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
678 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\v
679 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
680 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\ver
681 \end{itemize}
682
683 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
684 \begin{itemize}
685 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
686 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
687 \end{itemize}
688
689 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
690
691 \chapter{Alguns outros exemplo úteis}
692
693 \begin{tcolorbox}[title=Meu Textbox]
694 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
695 \end{tcolorbox}
696
697 \begin{tcolorbox}
698 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
699 \end{tcolorbox}
700
701 \begin{figure}[ht]
702 \centering
703 \begin{tikzpicture}
704 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{
705 \begin{scope}[x={(image.south east)},y={(image.north west)}]
706 % Definindo o textbox dentro da figura

```

```

707         \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text
708             \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
709                 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
710             \end{tcolorbox}
711         };
712     \end{scope}
713 \end{tikzpicture}
714 \caption{Figura com Textbox}
715 \label{fig:figura_com_textbox1}
716 \fonte{Elabora\c{c}\~ao pr\'opria.}
717 \end{figure}
718
719
720 \begin{figure}[ht]
721     \centering
722     \begin{tcolorbox}
723 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
724 \end{tcolorbox}
725     \caption{Figura com Textbox simples}
726     \label{fig:figura_com_textbox}
727     \fonte{Elabora\c{c}\~ao pr\'opria.}
728 \end{figure}
729
730 \section{Listagens de código e algoritmos}
731
732 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
733
734 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
735 def saudacao(nome):
736     # imprime uma saudação acentuada
737     print(f"Olá, {nome}!")
738
739 saudacao("COPPE")
740 \end{python}
741 \source{Elaboração própria.}
742
743 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
744
745 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
746 Olá, COPPE!
747 \end{Verbatim}
748
749 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
750 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
751 texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al
752
753 \begin{algorithm}[H]
754 \caption{Soma dos elementos de um vetor}
755 \label{alg:soma}
756 \Dados{vetor $v$ com $n$ elementos}
757 \Resultado{soma $s$ dos elementos de $v$}
758 $s \leftarrow 0$;
759 \Para{$i \leftarrow 1$ \Ate{} $n$}{
760     $s \leftarrow s + v[i]$;

```

```

761 }
762 \Retorna $$\;
763 \end{algorithm}
764
765 \chapter{Escrevendo matemática}
766
767 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carrega
768
769 \section{Alinhamento (align)}
770 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebrando
771 \begin{align}
772   f(x) &= (x+1)^2 \\
773         &= x^2 + 2x + 1 .
774 \end{align}
775
776 \section{Matrizes (matrix)}
777 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
778 \begin{equation}
779   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
780   \quad
781   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
782 \end{equation}
783
784 \section{Subequações (subequations)}
785 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
786 \begin{subequations}
787 \begin{align}
788   a &= b + c , \\
789   d &= e - f .
790 \end{align}
791 \end{subequations}
792
793 \section{Estruturas alinhadas (array)}
794 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
795 \begin{equation}
796   |x| = \left\{ \begin{array}{l}
797     x, \quad \text{se } x \geq 0 , \\
798     -x, \quad \text{caso contrário.}
799   \end{array} \right.
800 \end{equation}
801
802 \section{Operadores (\textbackslash DeclareMathOperator)}
803 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no preâmbulo
804 \begin{verbatim}
805 \usepackage{amsmath}
806 \DeclareMathOperator{\RMSE}{RMSE}
807 \DeclareMathOperator{\diag}{diag}
808 \end{verbatim}
809 Assim:
810 \begin{itemize}
811   \item \verb|\$ \RMSE(y, \hat{y})$| produz  $\text{RMSE}(y, \hat{y})$ ;
812   \item \verb|\$ \diag(\lambda_1, \lambda_2)$| produz
813          $\text{diag}(\lambda_1, \lambda_2)$ .
814 \end{itemize}

```

```

815
816 \section{Unicode e LuaLaTeX (unicode-math)}
817 \label{sec:unicode-math}
818 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss}
819 \begin{verbatim}
820 \usepackage{amsmath}      % a classe já carrega
821 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
822 \setmathfont{Latin Modern Math} % ou STIX Two Math
823 \end{verbatim}
824 O código clássico continua valendo ( $\mathbb{R}$ ),  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
825
826 \chapter{Referências cruzadas, notas e índices}
827 \label{cap:refs}
828
829 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
830
831 \section{Rótulos e referências cruzadas}
832
833 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
834
835 \section{Notas de rodapé}
836
837 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
838
839 \section{hyperref e marcadores}
840
841 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
842
843 \section{URLs}
844
845 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
846 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
847
848 \section{Índice remissivo}
849
850 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
851
852 \section{Traduções (NBR 10520)}
853
854 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
855
856 \chapter{Truques de \LaTeX}
857
858 \section{Comentários mágicos (\texttt{\% !TeX})}
859 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
860 \begin{verbatim}
861 % !TeX program = lualatex
862 % !TeX encoding = UTF-8
863 % !TeX spellcheck = pt_BR
864 % !BIB program = biber
865 \end{verbatim}
866 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
867
868 \section{Espaço engolido}

```



```

869 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como ‘‘\LaTeX é
870
871 \section{Escopo (grupos)}
872 Chaves \verb|{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
873
874 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash m
875 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
876
877 \section{\texttt{latexmkrc}, perl e Overleaf}
878 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
879
880 \chapter{Método Proposto}
881 \chapter{Resultados e Discuss\~oes}
882
883 \section{Algumas Demonstra\c c\~oes}
884
885 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
886 \syml[beta]{Beta}{A palavra Beta more e corrigida}
887 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
888 \syml[beta]{A palavra beta}
889 \syml[alpha]{A palavra alpha}
890 \syml[alpha]{Alpha}{A palavra Alpha}
891 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
892 \syml[marco]{Marco}{A palavra Marco corrigida}
893
894 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
895 \abbrev{GoT}{Game of Thrones}
896 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
897 \abbrev[iot]{IoT}{IoT ordenado como iot}
898 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
899 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
900 \abbrev{IoT}{IoT com ordenação default}
901 \abbrev[ITU]{ITU}{ITU mesmo}
902
903 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
904 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
905
906 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
907
908
909
910
911 \chapter{Conclusões}
912
913 \backmatter
914
915 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
916 % partitura --- declarados via sinônimos em português em example.bib.
917 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
918 \nocite{jurisprudencia-example}
919 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
920 \nocite{jornalcaderno-example,jornal-example}
921 \printbibliography
922

```

```

923
924
925 \appendix
926
927 \chapter{Um apêndice}
928
929 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
930
931 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
932
933
934 \chapter{Outro apêndice}
935
936 \annex
937
938
939 \chapter{Um Anexo}
940 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
941
942
943
944 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
945
946 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comando
947
948
949 \chapter{Outro Anexo}
950
951
952 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
953
954 \end{document}
955
956 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

957 \class
958 \def\filename{coppe.dtx}
959 \def\fileversion{v4.1}
960 \def\filedate{2026/09/09}
961 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
962 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
963 % Save the class version and date into private macros immediately, before any
964 % required package can redefine the very common \fileversion / \filedate
965 % global names (LaTeX has no convention reserving them for the host class).
966 % \coppe@version / \coppe@date are what user-facing macros (e.g.
967 % \coppetexfinalpage) should reference.
968 \let\coppe@version\fileversion
969 \let\coppe@date\filedate
970 % Document date stored in private slots so \date can set them without ever
971 % touching TeX's \month/\year primitives (which would break \today). They

```

```

972 % default to the current system month/year, so a document without \date
973 % still renders sensibly on the cover and folha de rosto.
974 \edef\coppe@docmonth{\the\month}
975 \edef\coppe@docyear{\the\year}
976 \LoadClass[12pt,a4paper,oneside]{book}
977 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
978 % the END of the class (after every class hook is defined), in that order:
979 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
980 % it patches (TOC, refs, lists) are already defined, and biblatex must come
981 % after hyperref so its citation links and back-references work.
982 \RequirePackage{hyphenat}
983 \RequirePackage{lastpage}
984 \RequirePackage{ifthen}
985 \RequirePackage{graphicx}
986 \RequirePackage{tabularx}
987 \RequirePackage{booktabs}          % regras de tabela profissionais (toprule/midrule/bottomrule)
988 \RequirePackage{longtable}        % tabelas que quebram entre páginas
989 \RequirePackage{etoolbox}
990 \RequirePackage{ltxcmds}
991 \RequirePackage{hologo}           % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
992 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
993 \def\TeX{\hologo{TeX}}
994 \def\LaTeX{\hologo{LaTeX}}
995 \def\LaTeXe{\hologo{LaTeXe}}
996 \def\BibTeX{\hologo{BibTeX}}
997 \def\pdfLaTeX{\hologo{pdfLaTeX}}
998 \def\pdfTeX{\hologo{pdfTeX}}
999 \def\LuaLaTeX{\hologo{LuaLaTeX}}
1000 \def\XeLaTeX{\hologo{XeLaTeX}}
1001 \def\MiKTeX{\hologo{MiKTeX}}
1002 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1003 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1004 % files may use accented Latin characters (âéíóú) directly. Under pdfLaTeX we also
1005 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1006 \RequirePackage{iftex}
1007 \ifPDFTeX
1008   \RequirePackage[utf8]{inputenc}
1009   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1010     Although slower, you should try compiling with LuaLaTeX}}
1011 \fi
1012 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1013 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1014 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1015 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1016 % 2. The consequences are not cosmetic:
1017 %
1018 % * the text renders blurred at any zoom and prints poorly;
1019 % * copying and searching text in the PDF is unreliable;
1020 % * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1021 %   attainable with bitmap text.
1022 %
1023 % Latin Modern is the vector successor of Computer Modern, metrically and
1024 % visually near-identical, and ships with every current distribution, so the
1025 % page layout does not move.

```

```

1026 \RequirePackage{lmodern}
1027 \RequirePackage[T1]{fontenc}
1028 % Math support. amsmath is universal and works on every engine, so the class
1029 % brings it in unconditionally (it must precede hyperref, which is loaded at
1030 % the end of the class). amssymb is the legacy AMS symbol package, built
1031 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1032 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1033 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1034 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1035 % unicode-math (the class does not auto-load unicode-math because choosing
1036 % an OpenType math font is a typographic decision that belongs to the author).
1037 \RequirePackage{amsmath}
1038 \ifPDFTeX
1039   \RequirePackage{amssymb}
1040 \else
1041   % Unicode engine: defer the check to end of document. Only warn if the
1042   % user did NOT add either unicode-math (the modern OpenType-math route)
1043   % or amssymb (the legacy route, which they would have loaded manually).
1044   % This avoids nagging authors who already followed the documented
1045   % \usepackage{unicode-math} workflow.
1046   \AtEndDocument{%
1047     \@ifpackageloaded{unicode-math}{}%
1048     \@ifpackageloaded{amssymb}{}%
1049     \ClassWarningNoLine{coppe}{%
1050       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1051       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1052       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1053       Add to your preamble:\MessageBreak
1054       \space\space\string\usepackage{unicode-math}\MessageBreak
1055       \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1056       Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1057       with the OpenType math font setup}}}%
1058 \fi
1059 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1060 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1061 % margins (right 3cm / left 2cm) and recommended printing the body on both
1062 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1063 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1064 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1065 % bindingoffset) no longer has any normative basis and is now opt-in via the
1066 % \texttt{twoside} class option, handled after \ProcessOptions below.
1067 %
1068 % headheight=15pt is given here (not patched later) so geometry's own page
1069 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1070 % height to host the running head, and the standard 12pt default triggers
1071 % its "headheight is too small" warning. Letting geometry know up front
1072 % keeps \topmargin a derived value -- we never touch it by hand.
1073 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1074 % absent it follows the class, which \LoadClass set to one-sided above.)
1075 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1076 % measures the right margin -- to the digits themselves: "a 2cm da borda
1077 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1078 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1079 % geometry derives \topmargin from top - 1in - headheight - headsep, so

```

```

1080 % shortening headsep lowers the header and leaves the body top untouched:
1081 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1082 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1083 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1084 % not put the digits flush with the top of \headheight. Two measured points,
1085 % taken from the rendered page rather than from the PDF's font metrics:
1086 % headsep 18.07pt -> digits at 1.8669cm headsep 14.29pt -> 2.0616cm
1087 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1088 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1089 % digits are shorter, and their top therefore sits lower for the same baseline:
1090 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1091 % precisely why this is a measured constant and not a computed one -- change
1092 % the folio's size or headheight and it has to be measured again, not scaled.
1093 % Verified on the rendered page: the top of the digits and their right edge come
1094 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1095 % shared 0.019cm is the ruler's own bias and not a residual error.
1096 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1097 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1098 % The first paragraph of every section must be indented like any other. LaTeX's
1099 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1100 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1101 % indentfirst was mentioned only as a commented suggestion in the example
1102 % preamble, so every document built with this class came out with the first
1103 % paragraph of each section flush left. Loading it in the class fixes it for
1104 % everybody instead of asking each author to remember.
1105 %
1106 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1107 % \@afterindentfalse.
1108 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1109 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1110 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1111 % which is the "text does not always respect the margins" a recent review
1112 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1113 % interword spaces instead of running into the margin. It cannot save a single
1114 % token longer than the measure, but it clears the ordinary cases.
1115 \setlength\emergencystretch{3em}
1116 \RequirePackage[indentfirst]
1117 \RequirePackage{titlesec}
1118 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1119 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capítulo N").
1120 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1121 % keep "N Title". \appendix and \annex raise the flag (see below).
1122 \newif\if@coppeappendix \@coppeappendixfalse
1123 \titleformat{\chapter}[hang]
1124 {\normalfont\Large\bfseries}
1125 {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1126 {1ex}\{\MakeUppercase}
1127 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1128 % 2.6 says the numeric indicative of a section precedes its title flush left,
1129 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1130 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1131 % indicative, and the manual sets its own Annex A centred. The chapter format is
1132 % therefore re-issued when \appendix or \annex is called: same label, same
1133 % travessao, but a centred block instead of a flush-left hang. Numbered chapters

```

```

1134 % are untouched -- they DO carry a numeric indicative and stay flush left.
1135 \newcommand\coppe@appendixheadings{%
1136   \titleformat{\chapter}[block]
1137     {\normalfont\Large\bfseries\filcenter}
1138     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1139     {1ex}{\MakeUppercase}%
1140   \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1141 \let\coppe@orig@appendix\appendix
1142 \renewcommand\appendix{%
1143   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1144 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1145 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1146 % which macro is written (so it is correct at .toc-render time even though the
1147 % appendixname has changed back); the chapter letter is baked in at write time.
1148 \newif\if@coppeannex \@coppeannexfalse
1149 \newcommand\coppe@tocapp[1]{%
1150   \MakeUppercase{\coppestring{appendix}}%
1151   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1152 \newcommand\coppe@tocanx[1]{%
1153   \MakeUppercase{\coppestring{annex}}%
1154   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1155 \def\@chapter[#1]#2{%
1156   \ifnum \c@secnumdepth >\m@ne
1157     \if@mainmatter
1158       \refstepcounter{chapter}%
1159       \typeout{\@chapapp\space\thechapter.}%
1160       \if@coppeappendix
1161         \if@coppeannex
1162           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1163         \else
1164           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1165         \fi
1166       \else
1167         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1168       \fi
1169     \else
1170       \addcontentsline{toc}{chapter}{#1}%
1171     \fi
1172   \else
1173     \addcontentsline{toc}{chapter}{#1}%
1174   \fi
1175   \chaptermark{#1}%
1176   \addtocontents{lof}{\protect\advspace{10\p@}}%
1177   \addtocontents{lot}{\protect\advspace{10\p@}}%
1178   \if@twocolumn
1179     \topnewpage[\@makechapterhead{#2}]%
1180   \else
1181     \@makechapterhead{#2}%
1182     \@afterheading
1183   \fi}
1184 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1185 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1186 \titleformat{name=\chapter,numberless}[hang]
1187   {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}

```

```

1188 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1189 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1190 % comment until v4.1, and made the secondary level as heavy as the primary one,
1191 % defeating the "destaque gradativo" of 2.6. The gradation is now
1192 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1193 % bold, subsubsection = Title Case bold italic, as read from the manual.
1194 \titleformat{\section}
1195   {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1196 % subsection: bold (author writes Title Case).
1197 \titleformat{\subsection}
1198   {\normalfont\bfseries}{\thesubsection}{1ex}{\}
1199 % subsubsection: bold italic.
1200 \titleformat{\subsubsection}
1201   {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\}
1202 % paragraph = quinary section: italic, the author capitalising only the first
1203 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1204 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1205 %
1206 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1207 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1208 % "after" spacing in \titlespacing turns it into a displayed heading.
1209 \titleformat{\paragraph}[hang]
1210   {\normalfont\itshape}{\theparagraph}{1ex}{\}
1211 \titlespacing*{\paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1212 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1213 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1214 % the numbering following the whole chain down to the quinary, and the manual's
1215 % own sumario is four levels deep, so tocdepth follows.
1216 \setcounter{secnumdepth}{4}
1217 \setcounter{tocdepth}{4}
1218 % --- 0 sumario (2.6 e 3.1.2.1.6) -----
1219 %
1220 % Duas exigencias o governam, e a segunda tem um exemplo obrigatorio.
1221 %
1222 % A 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no
1223 % corpo do trabalho". A gradacao do corpo -- primaria em CAIXA ALTA negrito,
1224 % secundaria em CAIXA ALTA, terciaria em negrito, quaternaria em negrito
1225 % italico, quinary em italico -- tem de reaparecer aqui. So as duas primeiras
1226 % reapareciam: da terciaria em diante o sumario saia em redondo, contradizendo
1227 % o corpo.
1228 %
1229 % A 3.1.2.1.6: os titulos "devem ser alinhados a esquerda... pela margem do
1230 % titulo do indicativo mais extenso", e "utilize como exemplo o sumario deste
1231 % trabalho". Medido no PDF do manual: o indicativo comeca em 3,00 cm nos cinco
1232 % niveis -- a margem -- e o titulo em 4,75 cm nos cinco. Uma coluna so.
1233 %
1234 % 0 \@dottedtocline do LaTeX faz o contrario. Recua progressivamente (3,00 /
1235 % 3,62 / 4,57 / 5,89 / 7,13 cm nesta classe) e da a cada nivel uma caixa de
1236 % indicativo de largura fixa. Isso empurrava o titulo quinary para 9,20 cm --
1237 % 6,2 dos 16 cm de mancha -- e, pior, a caixa fixa ESTOURA: com o indicativo
1238 % "10.10.10.10.10", que uma tese de dez capitulos alcanca, o numero entrava
1239 % 4,6 mm POR CIMA do titulo. A 2.6 quer o titulo "separado por um espaco" do
1240 % indicativo; ali nao havia espaco nenhum.
1241 %

```

```

1242 % A coluna abaixo e medida, nao arbitrada: \coppe@tocnumberline mede cada
1243 % indicativo enquanto compoe o sumario e guarda o maior; o valor vai para o
1244 % .aux e governa a passada seguinte. E literalmente "a margem do titulo do
1245 % indicativo mais extenso". Custa uma passada -- a mesma que as referencias
1246 % cruzadas ja custam.
1247 \newlength\coppe@tocnumwidth
1248 \newlength\coppe@tocnummax
1249 \newlength\coppe@tocnumsep
1250 % O espaco entre o indicativo e o titulo (2.6, "separado por um espaco").
1251 \setlength\coppe@tocnumsep{0.5em}
1252 % Piso: a coluna do proprio manual. Pela regra literal, um trabalho so com
1253 % capitulos teria uma coluna de meio centimetro e o titulo quase colado na
1254 % margem; o piso evita isso. A coluna nunca fica MENOR que o indicativo mais
1255 % extenso mais o espaco -- o piso so a impede de ficar estreita demais.
1256 \setlength\coppe@tocnumwidth{1.75cm}
1257 \setlength\coppe@tocnummax{\z@}
1258 \newcommand\coppe@tocnumberline[1]{%
1259   \settowidth\@tempdima{#1}%
1260   \ifdim\@tempdima>\coppe@tocnummax \global\coppe@tocnummax\@tempdima \fi
1261   \hb@xt@\coppe@tocnumwidth{#1\hfil}}
1262 % A largura medida na passada anterior chega pelo .aux, que o \document le
1263 % antes dos ganchos de \AtBeginDocument.
1264 \AtBeginDocument{%
1265   \@ifundefined{coppe@tocnumsaved}{}{%
1266     \setlength\@tempdima{\coppe@tocnumsaved}%
1267     \addtolength\@tempdima\coppe@tocnumsep
1268     \ifdim\@tempdima>\coppe@tocnumwidth
1269       \setlength\coppe@tocnumwidth\@tempdima
1270     \fi}}
1271 \AtEndDocument{%
1272   \ifdim\coppe@tocnummax>\z@
1273     \if@filesw
1274       \immediate\write\@auxout{\string\gdef\string\coppe@tocnumsaved
1275         {\the\coppe@tocnummax}}%
1276     \fi
1277     \setlength\@tempdima\coppe@tocnummax
1278     \addtolength\@tempdima\coppe@tocnumsep
1279     \ifdim\@tempdima>\coppe@tocnumwidth
1280       \ClassWarning{coppe}{A coluna do sumario cresceu; rode LaTeX de novo}%
1281     \fi
1282   \fi}
1283 % Uma linha do sumario. #1 nivel, #2 fonte, #3 entrada, #4 folha. A entrada
1284 % traz \numberline{N} seguido do titulo; o \hskip-\leftskip puxa a primeira
1285 % linha de volta a margem, e o \leftskip deixa as linhas seguintes de um
1286 % titulo longo alinhadas pela coluna do titulo -- nunca debaixo do numero.
1287 % A folha herda a fonte do nivel, como no sumario do manual.
1288 \newcommand\coppe@tocline[4]{%
1289   \ifnum #1>\c@tocdepth\else
1290     \vskip\z@\@plus.2\p@
1291     {\leftskip\z@ \rightskip\@tocrmarg \parfillskip-\rightskip
1292     \parindent\z@ \interlinepenalty\@M
1293     \leavevmode
1294     #2%
1295     \advance\leftskip\coppe@tocnumwidth

```



```

1296 \null\nobreak\hskip-\leftskip
1297 #3\nobreak
1298 \leaders\hbox{$\m@th\mkern\@dotsep mu\hbox{.}\mkern\@dotsep mu$}%
1299 \hfill\nobreak
1300 \hb@xt@\@pnumwidth{\hss #4}\par}%
1301 \fi}
1302 % 0 pontilhado do manual e mais fechado que o do book; \@dotsep e a distancia
1303 % entre os pontos, em unidades matematicas.
1304 \renewcommand\@dotsep{2}
1305 % Os cinco niveis. 0 \MakeUppercase vai so nos dois primeiros porque so eles
1306 % saem em caixa alta no corpo; ele atravessa os algarismos sem toca-los e
1307 % preserva comandos robustos, inclusive os do hyperref.
1308 %
1309 % 0 nivel 1 nao ganha mais o \addvspace{1.0em} do book: o sumario do manual
1310 % tem espacamento uniforme, sem folga extra antes da secao primaria. As
1311 % penalidades ficam, para que uma entrada de capitulo nao seja abandonada
1312 % sozinha no pe da folha.
1313 \renewcommand*\l@chapter[2]{%
1314 \addpenalty{-\@highpenalty}%
1315 \coppe@tocline{0}{\bfseries}{\MakeUppercase{#1}}{#2}%
1316 \penalty\@highpenalty}
1317 \renewcommand*\l@section[2]{%
1318 \coppe@tocline{1}{\normalfont}{\MakeUppercase{#1}}{#2}}
1319 \renewcommand*\l@subsection[2]{%
1320 \coppe@tocline{2}{\normalfont\bfseries}{#1}{#2}}
1321 \renewcommand*\l@subsubsection[2]{%
1322 \coppe@tocline{3}{\normalfont\bfseries\itshape}{#1}{#2}}
1323 \renewcommand*\l@paragraph[2]{%
1324 \coppe@tocline{4}{\normalfont\itshape}{#1}{#2}}
1325 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1326 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1327 \RequirePackage{caption}
1328 % justification=centering makes MULTI-LINE captions centre like the one-line
1329 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1330 % line and justifies one that wraps, so in a document with mixed caption lengths
1331 % a couple of tables looked left-aligned while the rest looked centred -- which
1332 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1333 % prescribe the alignment; it does not tolerate two alignments either.
1334 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1335 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1336 % caption so it picks up the ABNT captionsetup above.
1337 \RequirePackage{subcaption}
1338 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1339 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1340 % at the bottom is mandatory). The float package's plaintop style forces the
1341 % caption above the body for figure, table and the new quadro float. -----
1342 \RequirePackage{float}
1343 \floatstyle{plaintop}
1344 \restylefloat{figure}
1345 \restylefloat{table}
1346 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1347 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1348 % unlike a tabela (rules on top and bottom only).
1349 \providecommand{\quadroname}{\coppestring{frame}}

```

```

1350 \providecommand{\listquadroname}{\coppestring{listframe}}
1351 \newfloat{quadro}{htbp}{loq}[chapter]
1352 \floatname{quadro}{\quadroname}
1353 \providecommand{\quadroautorefname}{\quadroname}
1354 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1355 % (float's own \l@quadro lays the number/title out incorrectly).
1356 \let\l@quadro\l@figure
1357 \newcommand\listofquadros{%
1358   \chapter*{\listquadroname}%
1359   \coppe@listtoc{\listquadroname}%
1360   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1361   \@starttoc{loq}}
1362 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1363 % float as a smaller, single-spaced, centred line; it does NOT advance any
1364 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1365 % the label is "Fonte" (pt) / "Source" (en).
1366 \newcommand{\cpsourcenamename}{\coppestring{source}}
1367 \newcommand{\cpsource}[1]{%
1368   \par\addvspace{\smallskipamount}%
1369   {\footnotesize\singlespacing\centering\cpsourcenamename: #1\par}}
1370 \let\cpfonte\cpsource
1371 \ifundefined{source}{\let\source\cpsource}{}%
1372 \ifundefined{fonte}{\let\fonte\cpsource}{}%
1373 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1374 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1375 \newcommand{\newcoppefloat}[3]{%
1376   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1377   \floatname{#1}{#2}%
1378   \expandafter\newcommand\csname listof#1s\endcsname{%
1379     \chapter*{#3}\coppe@listtoc{#3}%
1380     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1381     \@starttoc{lo#1}}}
1382 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1383 % (filete) from the left margin; footnotesize is the book default. -----
1384 \let\coppe@orig@footnotetext\@footnotetext
1385 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1386 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1387 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1388 % (manual2024 2.6 / proposta P24). -----
1389 \RequirePackage{enumitem}
1390 \newlist{alineas}{enumerate}{2}
1391 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1392 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1393 % --- Page numbering: arabic in the top OUTER corner on textual pages
1394 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1395 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1396 % class option). -----
1397 % Every list goes through \@starttoc, which allocates a stream the first time
1398 % it sees an extension. That is where the ceiling is hit in practice, so that
1399 % is where the class checks -- one \ifnum before the allocation, and an error
1400 % that names the three ways out instead of the kernel's bare "No room".
1401 %
1402 % \count17 is the write allocation counter, and 15 is the last usable stream.
1403 % Under LuaTeX there are 128 and the counter is managed differently, so the

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1404 % check is skipped there; the same is true when morewrites is in charge.
1405 % The wrap waits for \begin{document} on purpose: hyperref redefines
1406 % \starttoc as well, and it is loaded at the very end of this class, so a
1407 % \let taken here would be thrown away.
1408 \newcommand\coppe@checkwrites[1]{%
1409   \ifdefined\directlua\else
1410     \if@coppemorewrites\else
1411       \@ifundefined{tf@#1}{%
1412         \ifnum\count17>14
1413           \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1414             Este trabalho pede mais listas do que o pdfTeX consegue manter
1415             abertas ao mesmo tempo.\MessageBreak
1416             Tres saidas, nesta ordem de preferencia:\MessageBreak
1417             (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1418             (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1419             (3) remova uma lista que o trabalho nao usa.\MessageBreak
1420             A lista que faltou agora foi a de extensao '#1', mas o problema
1421             nao e dela: e a soma de todas.}%
1422         \fi}{}%
1423     \fi
1424   \fi}
1425 % The wrapping itself lives in a macro of its own rather than inside the hook:
1426 % \AtBeginDocument stores its argument by expansion, and a parameter written
1427 % there does not survive it -- the document died with "Parameters must be
1428 % numbered consecutively" at \begin{document}. One level of nesting, one
1429 % doubling, and the hook only has to call it.
1430 \newcommand\coppe@wrapstarttoc{%
1431   \let\coppe@orig@starttoc\@starttoc
1432   % \numberline mede o indicativo (ver \coppe@tocnumberline) apenas no sumario:
1433   % a lista de figuras e a de tabelas numeram ilustracoes, nao secoes, e nao tem
1434   % por que compartilhar a coluna.
1435   \def\@starttoc##1{%
1436     \coppe@checkwrites{##1}%
1437     \begingroup
1438       \def\coppe@@ext{##1}\def\coppe@@toc{toc}%
1439       \ifx\coppe@@ext\coppe@@toc \let\numberline\coppe@tocnumberline \fi
1440       \coppe@orig@starttoc{##1}%
1441     \endgroup}
1442 \AtBeginDocument{\coppe@wrapstarttoc}
1443 \RequirePackage{fancyhdr}
1444 % \headheight is set to 15pt by geometry up above (see the geometry call),
1445 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1446 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1447 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1448 % folio was the one item on that list still coming out at the body size of 12pt.
1449 % It is \footnotesize now, which is the same smaller size the other three
1450 % already use -- "uniforme" is part of the requirement, not just "menor".
1451 % Pre-textual lists reach the sumario only under the listasnosumario option
1452 % (see the option's declaration for the reason). Post-textual material calls
1453 % \addcontentsline directly and is unaffected.
1454 \newcommand\coppe@listtoc[1]{%
1455   \if@coppelistastoc\addcontentsline{toc}{chapter}{#1}\fi
1456   \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1457   % the list of abbreviations and the list of symbols -- apart from the genuine

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```

1458 % post-textual Glossario, which shares the same environment but must appear in
1459 % the sumario.
1460 \newif\if@coppeglosspre \@coppeglossprefalse
1461 \newcommand\coppe@folio{{\footnotesize\thepage}}
1462 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}}%
1463 \renewcommand{\headrulewidth}{0pt}}
1464 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1465 % number may leak onto them (e.g. the verso right after the cover).
1466 \renewcommand{\cleardoublepage}{%
1467 \clearpage
1468 \if@twoside
1469 \ifodd\c@page\else
1470 \hbox{}\thispagestyle{empty}\newpage
1471 \if@twocolumn\hbox{}\newpage\fi
1472 \fi
1473 \fi}
1474 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1475 p\kern-.025em e}}\kern-.08em
1476 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

1477 \newboolean{maledoc}
1478 \setboolean{maledoc}{false}
1479 % maledoc diz apenas que o substantivo do tipo de documento e masculino
1480 % ("Exame de Qualificacao ... apresentado", "SUBMETIDO"): serve a concordancia,
1481 % e nada mais. Quem NAO vai a deposito -- e portanto nao tem ficha
1482 % catalografica nem folha da Coleta CAPES -- e o exame de qualificacao, que e
1483 % outra coisa. As duas condicoes coincidem hoje por acaso, e por isso ganham
1484 % cada uma o seu booleano.
1485 \newif\if@coppeexame \@coppeexamefalse
1486 %
1487 % Class options.
1488 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1489 % fixed language slots:
1490 % * main -- the language of the body, captions, and the main abstract.
1491 % Set by a class option: brazilian (default), english,
1492 % spanish, french, italian, or any user-supplied language
1493 % that ships a coppe-lang-<name>.def file.
1494 % * foreign -- the language used by the foreignabstract environment and
1495 % by \foreign@... registers. Defaults to english (or to
1496 % brazilian when main=english) for back-compat.
1497 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1498 % in the preamble (for citations or quotations in a third
1499 % language). Has no class option of its own.
1500 % Note: the legacy \if@english boolean (defined and toggled here in
1501 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1502 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.lbr)
1503 % tests it. Code that needs to branch on the main language should test
1504 % \ifx\coppe@mainlang\coppe@englishstr instead.
1505 \def\coppe@mainlang{brazilian}
1506 \def\coppe@foreignlang{english}
1507 \DeclareOption{brazilian}{%
1508 \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1509 \DeclareOption{english}{%
1510 \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1511 \DeclareOption{spanish}{%

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1512 \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1513 \DeclareOption{french}{%
1514 \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1515 \DeclareOption{italian}{%
1516 \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1517 \DeclareOption{msc}{%
1518 \newcommand{\@degree}{M.Sc.}
1519 \newcommand{\@degreename}{Mestrado}
1520 \newcommand{\local@degname}{Mestre}
1521 \newcommand{\foreign@degname}{Master}
1522 \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1523 \newcommand\foreign@doctype{Dissertation}
1524 }
1525 \DeclareOption{dsccexam}{%
1526 \newcommand{\@degree}{D.Sc.}
1527 \newcommand{\@degreename}{Doutorado}
1528 \newcommand{\local@degname}{Doutor}
1529 \newcommand{\foreign@degname}{Doctor}
1530 \setboolean{maledoc}{true}\@coppeexamtrue
1531 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1532 \newcommand\foreign@doctype{Qualifying Exam}
1533 }
1534 \DeclareOption{mscexam}{%
1535 \newcommand{\@degree}{M.Sc.}
1536 \newcommand{\@degreename}{Mestrado}
1537 \newcommand{\local@degname}{Mestre}
1538 \newcommand{\foreign@degname}{Master}
1539 \setboolean{maledoc}{true}\@coppeexamtrue
1540 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1541 \newcommand\foreign@doctype{Qualifying Exam}
1542 }
1543 \DeclareOption{dsc}{%
1544 \newcommand{\@degree}{D.Sc.}
1545 \newcommand{\@degreename}{Doutorado}
1546 \newcommand{\local@degname}{Doutor}
1547 \newcommand{\foreign@degname}{Doctor}
1548 \newcommand\local@doctype{Tese}
1549 \newcommand\foreign@doctype{Thesis}
1550 }
1551 \newif\if@coppenumeric\@coppenumericfalse
1552 \DeclareOption{numbers}{\@coppenumerictrue}
1553 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1554 % pre-textual pages are counted but not numbered (manual2024 2.7).
1555 \newif\if@copperoman\@copperomanfalse
1556 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1557 \newif\if@coppedoublespace\@coppedoublespacefalse
1558 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1559 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1560 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},

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1561 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1562 % With no \fichacatalografica file given, the folha adicional leaves a framed
1563 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1564 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1565 % for deposit: the official ficha comes from the SiBI generator.
1566 % Annex D of the manual draws one signature rule per board member, with the
1567 % identification underneath. The COPPE folha de aprovacao has listed the names
1568 % compactly, without rules, for years. The compact list stays the default and
1569 % the rules are opt-in, so existing theses do not change shape overnight.
1570 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1571 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1572 % encryption, restricted transparency) can trip up a document that pulls in
1573 % arbitrary images or packages, and breaking existing theses to enforce it would
1574 % be worse than letting the author switch it on when depositing.
1575 \newif\if@coppepdfa \@coppepdfafalse
1576 \DeclareOption{pdfa}{\@coppepdfatrue}
1577 \newif\if@coppeassin \@coppeassinfalse
1578 \DeclareOption{assinaturas}{\@coppeassintrue}
1579 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1580 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1581 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1582 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1583 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.
1584 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1585 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1586 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1587 % em que a materia nele se sucede" and adds "inclusive os elementos
1588 % pos-textuais" -- and then settles the question by pointing at its own sumario
1589 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1590 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1591 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1592 % The class was adding all of them, so every thesis carried six entries the
1593 % manual's own example does not have. They come out. Post-textual elements --
1594 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1595 %
1596 % Programas that want the old behaviour ask for it by name.
1597 \newif\if@coppelistastoc \@coppelistastocfalse
1598 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1599 \newif\if@copperascunho \@copperascunhofalse
1600 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1601 \newif\if@coppetwoaside \@coppetwoasidefalse
1602 \DeclareOption{twoside}{\@coppetwoasidetrue}
1603 \DeclareOption{oneside}{\@coppetwoasidefalse}
1604 % --- Fluxos de escrita -----
1605 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1606 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1607 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1608 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1609 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1610 % A work that asks for all of them sits exactly at the ceiling.
1611 %
1612 % The kernel's failure in that situation -- "No room for a new \write" -- names
1613 % neither the cause nor the way out, and lands on whichever list happens to be
1614 % requested last, which is usually \tableofcontents and has nothing to do with

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1615 % it. The class therefore checks before the allocation and says what is
1616 % happening (see \coppe@starttoc below).
1617 %
1618 % There are three ways out and the class chooses none of them for the author:
1619 %
1620 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1621 % and the class supports it.
1622 % * let the class load 'morewrites', which multiplexes the streams so the
1623 % ceiling disappears under pdfLaTeX.
1624 % * drop a list the work does not use. Each one returns a stream.
1625 %
1626 % The DEFAULT is the second one, and it is on. An error telling the author to
1627 % pass an option is advice for someone who already knows: a student two days
1628 % from the deposit reads "No room for a new \write" and concludes the class is
1629 % broken. So under pdfTeX the class loads morewrites by itself.
1630 %
1631 % Three things keep that from becoming a burden:
1632 %
1633 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it
1634 % and warns; morewrites does not become a hard dependency of a work that
1635 % would never have reached the ceiling anyway.
1636 % * 'semmorewrites' turns it off, for whoever hits a conflict. morewrites
1637 % patches \newwrite, \openout and \write and buffers what is written until
1638 % shipout, so a package that writes a file and reads it back in the SAME run
1639 % can misbehave. That is the real cost of this default, and it is why the
1640 % escape exists.
1641 % * 'morewrites' still exists and now means "I insist": with it, a missing
1642 % package is an error rather than a warning.
1643 %
1644 % Under LuaTeX none of this applies and nothing is loaded.
1645 \newif\if@coppemorewrites \@coppemorewritetrue
1646 \newif\if@coppemwforced \@coppemwforcedfalse
1647 \DeclareOption{morewrites}{\@coppemorewritetrue\@coppemwforcedtrue}
1648 \DeclareOption{semmorewrites}{\@coppemorewritetrue\@coppemwforcedfalse}
1649 % A 2.2(b) fixa a cor, o corpo 12 e o corpo menor uniforme dos quatro itens
1650 % listados nela, e nao diz uma palavra sobre familia de fonte -- nem a Norma da
1651 % COPPE. A escolha e do autor. O padrao da classe e a serifada, que e a do
1652 % LaTeX; 'semserifa' troca a familia padrao para a sem serifa. Note que o
1653 % manual do SiBI, que fixa a regra, esta ele proprio composto em Arial.
1654 \newif\if@coppesansserif \@coppesansseriffalse
1655 \DeclareOption{semserifa}{\@coppesansseriftrue}
1656 \ProcessOptions\relax
1657 % A troca da familia padrao vai aqui porque so agora se sabe se a opcao foi
1658 % pedida; o lmodern ja esta carregado la em cima e traz as duas familias, a
1659 % serifada e a sem serifa, ambas vetoriais -- que e o que o PDF/A exige.
1660 % Quem quiser uma sem serifa DIFERENTE (Helvetica, por exemplo) carrega o
1661 % pacote da fonte no preambulo e refaz \familydefault ali; esta opcao existe
1662 % para o caso comum, em que ninguem quer escolher fonte, so nao quer serifa.
1663 \if@coppesansserif
1664 \renewcommand{\familydefault}{\sfdefault}%
1665 \fi
1666 % morewrites has to come before anything else allocates, so it goes first
1667 % thing after the options are read.
1668 \if@coppemorewrites

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1669 \ifdefined\directlua
1670 % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1671 % cleared too, so the \@starttoc guard further down knows the ceiling is not
1672 % in play here either.
1673 \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1674 \@coppemorewritesfalse
1675 \else
1676 \IfFileExists{morewrites.sty}
1677 {\RequirePackage{morewrites}}
1678 {\if@coppemwforced
1679 \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1680 esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1681 que nao precisa dele.}%
1682 \else
1683 \ClassWarningNoLine{coppe}{O pacote morewrites nao esta instalado.\MessageBreak
1684 Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1685 com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak
1686 de fluxo, instale morewrites ou compile com lualatex}%
1687 \@coppemorewritesfalse
1688 \fi}%
1689 \fi
1690 \fi
1691 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1692 % two-sided layout can only be applied here. geometry reads left/right as
1693 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1694 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1695 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1696 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[R0,LE]
1697 % declarations elsewhere already put the folio in the top right corner of every
1698 % folha, as 2.7 requires.
1699 \if@coppetwoside
1700 \@twosidetrue
1701 \geometry{twoside}%
1702 \fi
1703 % Built-in language names (used by the loader and the dispatcher below).
1704 \def\coppe@brazilianstr{brazilian}
1705 \def\coppe@englishstr{english}
1706 % Load Babel with the foreign language first and the main language last
1707 % (Babel treats the LAST option in the list as the main language). With
1708 % pt-main this produces [english,brazilian]{babel} and with en-main
1709 % [brazilian,english] -- identical to the historical \if@english calls.
1710 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1711 % Some Babel modules change LaTeX beyond translating captions, and one of
1712 % those changes breaks the norm. With Spanish as the MAIN language,
1713 % babel-spanish redefines \numberline to put a period after the section
1714 % number, so the sumario reads "2.1. SECCION SECUNDARIA" while the section
1715 % heading itself -- which the class formats -- reads "2.1 SECCION SECUNDARIA".
1716 % The 2.6 separates the indicative from the title by a space (NBR 6024) and
1717 % the 3.1.2.1.6 wants the sumario to reproduce the text; the period fails
1718 % both. 'es-nosectiondot' turns off that single adjustment and leaves the rest
1719 % of babel-spanish (quoting, shorthands, roman numerals, captions) alone.
1720 \coppe@babelextra@<lang> is the general hook: it applies to the MAIN
1721 % language only, which is where these layout modules act, and it is kept out
1722 % of \coppe@babelopts because the language-pack loader below iterates that

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1723 % list and would look for coppe-lang-es-nosectiondot.def.
1724 \@namedef{coppe@babelextra@spanish}{,es-nosectiondot}
1725 \let\coppe@babelload\coppe@babelopts
1726 \@ifundefined{coppe@babelextra@\coppe@mainlang}{}{%
1727   \edef\coppe@babelload{\coppe@babelload\@nameuse{coppe@babelextra@\coppe@mainlang}}
1728 \expandafter\RequirePackage\expandafter[\coppe@babelload]{babel}
1729 % --- Multilingual string dispatcher -----
1730 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1731 % language <lang>. The three retrievers below look strings up either at the
1732 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1733 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1734 \providecommand{\copperdefstring}[3]{%
1735   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1736 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language @#1}}
1737 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1738 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1739 % Load an extra language pack at any point in the preamble (third-language
1740 % slot for citations or quotations beyond the main+foreign pair).
1741 \providecommand{\usecoppelanguage}[1]{%
1742   \def\coppe@@cur{#1}%
1743   \ifx\coppe@@cur\coppe@@brazilianstr\else
1744   \ifx\coppe@@cur\coppe@@englishstr\else
1745     \InputIfFileExists{coppe-lang-#1.def}{}
1746     {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1747   \fi\fi}
1748 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1749 % consulted by the cover/folha renderers once they are migrated in step 2.
1750 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1751 % still drive every visible site, so behavior for pt/en is unchanged.
1752 \providecommand{\titlein}[2]{%
1753   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1754 % --- Built-in string tables: brazilian + english. The values mirror the
1755 % literals currently scattered through the class; step-1 only populates the
1756 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1757 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1758 \copperdefstring{english}{appendix}{Appendix}
1759 \copperdefstring{brazilian}{annex}{Anexo}
1760 \copperdefstring{english}{annex}{Annex}
1761 \copperdefstring{brazilian}{frame}{Quadro}
1762 \copperdefstring{english}{frame}{Frame}
1763 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1764 \copperdefstring{english}{listframe}{List of Frames}
1765 \copperdefstring{brazilian}{source}{Fonte}
1766 \copperdefstring{english}{source}{Source}
1767 \copperdefstring{brazilian}{program}{Programa}
1768 \copperdefstring{english}{program}{Program}
1769 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1770 \copperdefstring{english}{listprogram}{List of Programs}
1771 \copperdefstring{brazilian}{references}{Refer\~encias}
1772 \copperdefstring{english}{references}{References}
1773 \copperdefstring{brazilian}{in:}{em}
1774 \copperdefstring{english}{in:}{in}
1775 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1776 \copperdefstring{english}{art}{Art}

```

1777 \copperdefstring{brazilian}{platform} {Plataforma}
1778 \copperdefstring{english} {platform} {Platform}
1779 \copperdefstring{brazilian}{directedby} {Dire\c{c}\~ao}
1780 \copperdefstring{english} {directedby} {Directed by}
1781 \copperdefstring{brazilian}{producedby} {Produ\c{c}\~ao}
1782 \copperdefstring{english} {producedby} {Produced by}
1783 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1784 \copperdefstring{english} {listabbreviation}{List of Abbreviations and Acronyms}
1785 \copperdefstring{brazilian}{listsymbol} {Lista de S{\`i}mbolos}
1786 \copperdefstring{english} {listsymbol} {List of Symbols}
1787 \copperdefstring{brazilian}{glossary} {Gloss{\`a}rio}
1788 \copperdefstring{english} {glossary} {Glossary}
1789 \copperdefstring{brazilian}{advisor} {Orientador}
1790 \copperdefstring{english} {advisor} {Advisor}
1791 \copperdefstring{brazilian}{advisors} {Orientadores}
1792 \copperdefstring{english} {advisors} {Advisors}
1793 \copperdefstring{brazilian}{dept} {Programa}
1794 \copperdefstring{english} {dept} {Department}
1795 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1796 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1797 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1798 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1799 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1800 \copperdefstring{brazilian}{volumename} {Volume}
1801 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1802 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1803 \copperdefstring{brazilian}{ofname} {de}
1804 \copperdefstring{english} {approvedmale} {Approved by}
1805 \copperdefstring{english} {approvedfemale}{Approved by}
1806 \copperdefstring{english} {approvedonmale} {Approved on}
1807 \copperdefstring{english} {approvedonfemale}{Approved on}
1808 \copperdefstring{english} {keywordsname} {Keywords}
1809 \copperdefstring{english} {volumename} {Volume}
1810 \copperdefstring{english} {coadvisor} {Co-advisor}
1811 \copperdefstring{english} {coadvisors} {Co-advisors}
1812 \copperdefstring{english} {ofname} {of}
1813 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1814 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1815 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1816 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1817 \copperdefstring{english} {cityname} {Rio de Janeiro}
1818 \copperdefstring{brazilian}{statename} {RJ}
1819 \copperdefstring{english} {statename} {RJ}
1820 \copperdefstring{brazilian}{countryname} {Brasil}
1821 \copperdefstring{english} {countryname} {Brasil}
1822 % Month names.
1823 \copperdefstring{brazilian}{monthname1} {Janeiro}
1824 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1825 \copperdefstring{brazilian}{monthname3} {Mar\c{c}o}
1826 \copperdefstring{brazilian}{monthname4} {Abril}
1827 \copperdefstring{brazilian}{monthname5} {Maio}
1828 \copperdefstring{brazilian}{monthname6} {Junho}
1829 \copperdefstring{brazilian}{monthname7} {Julho}
1830 \copperdefstring{brazilian}{monthname8} {Agosto}

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1831 \copperdefstring{brazilian}{monthname9} {Setembro}
1832 \copperdefstring{brazilian}{monthname10}{Outubro}
1833 \copperdefstring{brazilian}{monthname11}{Novembro}
1834 \copperdefstring{brazilian}{monthname12}{Dezembro}
1835 \copperdefstring{english}{monthname1} {January}
1836 \copperdefstring{english}{monthname2} {February}
1837 \copperdefstring{english}{monthname3} {March}
1838 \copperdefstring{english}{monthname4} {April}
1839 \copperdefstring{english}{monthname5} {May}
1840 \copperdefstring{english}{monthname6} {June}
1841 \copperdefstring{english}{monthname7} {July}
1842 \copperdefstring{english}{monthname8} {August}
1843 \copperdefstring{english}{monthname9} {September}
1844 \copperdefstring{english}{monthname10}{October}
1845 \copperdefstring{english}{monthname11}{November}
1846 \copperdefstring{english}{monthname12}{December}
1847 % algorithm2e package option for this language (empty -> english default).
1848 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1849 \copperdefstring{english} {algorithm2eopt}{}
1850 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1851 \copperdefstring{brazilian}{babelname}{brazilian}
1852 \copperdefstring{english} {babelname}{english}
1853 % Auto-load language packs for any non-built-in language listed in babelopts.
1854 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1855 % (defined just above) and need no extra file; others (spanish/french/italian
1856 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1857 % must already be defined when these run, so the loader sits AFTER the tables.
1858 \@for\coppe@@lang:=\coppe@babelopts\do{%
1859   \edef\coppe@@cur{\coppe@@lang}%
1860   \ifx\coppe@@cur\coppe@@brazilianstr\else
1861     \ifx\coppe@@cur\coppe@@englishstr\else
1862       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1863       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1864       {\ClassError{coppe}
1865         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1866         {Install it next to coppe.cls, or pick a built-in language.}}%
1867     \fi\fi}
1868 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1869 % Portuguese alias for short, inline quotations (proposta P3).
1870 \RequirePackage{csquotes}
1871 \newcommand{\citacao}[1]{\enquote{#1}}
1872 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1873 % package gives the "Programa" listing (caption on top, ABNT R46) with
1874 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1875 % the language styles below are based on the user's listagens.tex. -----
1876 \RequirePackage{xcolor}
1877 \RequirePackage{listings}
1878 \RequirePackage{fancyvrb}
1879 \DeclareFixedFont{\ttb}{T1}{ttxtt}{bx}{n}{10}
1880 \DeclareFixedFont{\ttm}{T1}{ttxtt}{m}{n}{10}
1881 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1882   literate=%
1883   {ã}{\~{a}}1 {á}{\~{a}}1 {â}{\~{a}}1 {ä}{\~{a}}1 {é}{\~{e}}1%
1884   {ê}{\~{e}}1 {ë}{\~{e}}1 {î}{\~{i}}1 {í}{\~{i}}1 {ô}{\~{o}}1%

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1885 {\o}{\'}{o}}1 {\o}{\~{o}}1 {\u}{\~{u}}1 {\u}{\'}{u}}1 {\c}{\c{c}}1%
1886 {\C}{\c{C}}1 {\A}{\'}{A}}1 {\A}{\~{A}}1 {\A}{\'}{A}}1 {\E}{\'}{E}}1%
1887 {\E}{\~{E}}1 {\I}{\'}{I}}1 {\O}{\'}{O}}1 {\O}{\~{O}}1 {\O}{\~{O}}1%
1888 {\U}{\'}{U}}1%
1889 }
1890 % Babel-aware listing names, set after listings' own babel setup.
1891 \AtBeginDocument{%
1892 \renewcommand\lstlistingname{\coppestring{program}}%
1893 \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1894 % \listofprogramas in the coppe list style (heading + ToC + running head).
1895 \newcommand\listofprogramas{%
1896 \chapter*{\lstlistlistingname}%
1897 \coppe@listtoc{\lstlistlistingname}%
1898 \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1899 \@starttoc{lol}}
1900 % Per-language styles + environments (python, java, xml, html, prolog).
1901 % The red hook that marks a wrapped line is built ONCE, into a box, and the
1902 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
1903 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
1904 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
1905 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
1906 % from inside listings' line-breaking machinery then fails with "Missing
1907 % number, treated as zero" at the first wrapped line. Every listing in the
1908 % document dies, so the class's own example could not be compiled as PDF/A at
1909 % all. A box is immune: it is built here, in ordinary horizontal mode where
1910 % xcolor works normally, and listings only ever copies it.
1911 %
1912 % The \sbox waits for \begin{document} because xcolor's colours are not
1913 % defined while the class is still loading.
1914 \newsavebox\coppe@lstarrow
1915 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
1916 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1917 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1918 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1919 emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1920 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1921 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1922 escapechar=',postbreak=\usebox\coppe@lstarrow}}
1923 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1924 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1925 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}%
1926 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}%
1927 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1928 numberstyle=\small\ttfamily,numbersep=0.5em,%
1929 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1930 keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1931 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1932 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1933 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1934 postbreak=\usebox\coppe@lstarrow}}
1935 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}}%
1936 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1937 numberstyle=\small\ttfamily,numbersep=0.5em,%
1938 keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%

```

```

1939 stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
1940 extendedchars=true,inputencoding=utf8,breaklines=true,%
1941 postbreak=\usebox\coppe@lstarrow}}
1942 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}}{
1943 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1944 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1945 keywordstyle=\ttb\color[rgb]{0,0,0.5},emph={MyClass,__init__},%
1946 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1947 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1948 breaklines=true,escapechar=',%
1949 postbreak=\usebox\coppe@lstarrow}}
1950 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}%
1951 \curname\@lst @SetFirstNumber\endcsname}{\curname\@lst @SaveFirstNumber\endcsname}
1952 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1953 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1954 morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
1955 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1956 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1957 breaklines=true,escapechar=',%
1958 postbreak=\usebox\coppe@lstarrow}}
1959 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}%
1960 \curname\@lst @SetFirstNumber\endcsname}{\curname\@lst @SaveFirstNumber\endcsname}
1961 \newcommand\javainline[1][\{\javastyle\lstinline!#1!\}}
1962 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}\}}
1963 % fancyvrb's Verbatim environment is available for program output; the
1964 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1965 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1966 % example,
1967 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1968 % ...
1969 % \end{Verbatim}
1970 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1971 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1972 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1973 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1974 % language (English) -- needed because algorithm2e errors on an empty option.
1975 \edef\coppe@algotlang{\coppemainstring{algorithm2eopt}}
1976 \ifx\coppe@algotlang\empty
1977 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1978 \else
1979 \expandafter\RequirePackage\expandafter[\coppe@algotlang,linesnumbered,lined,algochapter,r
1980 \fi
1981 % algorithm2e composes its own caption and therefore ignores the class-wide
1982 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
1983 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
1984 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
1985 % designating word, the number, a TRAVESSAO and the title, uniformly.
1986 %
1987 % The separator and the caption fonts are therefore set to match. The 'ruled'
1988 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
1989 % the source go, not whether the float is delimited, and the rules read here
1990 % like the top-and-bottom delimitation a table carries.
1991 \SetAlgoCaptionSeparator{\textendash}
1992 \SetAlCapFnt{\footnotesize}

```

```

1993 \SetAlCapNameFnt{\footnotesize}
1994 \SetAlCapSty{textnormal}
1995 \SetAlCapNameSty{textnormal}
1996 \SetKwFor{Para}{para}{faça}{fim do para}
1997 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1998 % setspace + hyperref + biblatex are loaded at the very end of the class
1999 % (see the block just before %</class>), so their hooks see every macro the
2000 % class defines, and the load order setspace -> hyperref -> biblatex is
2001 % respected.

\department This macro is used to set the author's affiliation. There are twelve options which
correspond to all academic units at COPPE/UFRJ. It defines the current and the
foreign names of these units.

2002 \newcommand\coppe@deptcode{}
2003 \newcommand\meta@deptname{}
2004 \newcommand\department[1]{%
2005 % The programme CODE is kept alongside the resolved name, and each name is
2006 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
2007 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
2008 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
2009 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
2010 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
2011 % subject line used to fall back to the bare programme code. Both forms are
2012 % written out here rather than derived from one another because there is no
2013 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
2014 % written, and a table of thirteen entries is cheaper than that machinery.
2015 \gdef\coppe@deptcode{#1}%
2016 \ifthenelse{\equal{#1}{PEB}}{
2017   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
2018   \global\def\meta@deptname{Engenharia Biomédica}
2019   \global\def\foreign@deptname{Biomedical Engineering}}{}
2020 \ifthenelse{\equal{#1}{PEC}}{
2021   {\global\def\local@deptname{Engenharia Civil}
2022   \global\def\meta@deptname{Engenharia Civil}
2023   \global\def\foreign@deptname{Civil Engineering}}{}
2024 \ifthenelse{\equal{#1}{PEE}}{
2025   {\global\def\local@deptname{Engenharia El{\` e}trica}
2026   \global\def\meta@deptname{Engenharia Elétrica}
2027   \global\def\foreign@deptname{Electrical Engineering}}{}
2028 \ifthenelse{\equal{#1}{PEM}}{
2029   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
2030   \global\def\meta@deptname{Engenharia Mecânica}
2031   \global\def\foreign@deptname{Mechanical Engineering}}{}
2032 \ifthenelse{\equal{#1}{PEMM}}{
2033   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
2034   \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
2035 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
2036 \ifthenelse{\equal{#1}{PEN}}{
2037   {\global\def\local@deptname{Engenharia Nuclear}
2038   \global\def\meta@deptname{Engenharia Nuclear}
2039   \global\def\foreign@deptname{Nuclear Engineering}}{}
2040 \ifthenelse{\equal{#1}{PEN0}}{
2041   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
2042   \global\def\meta@deptname{Engenharia Oceânica}

```

```

2043 \global\def\foreign@deptname{Ocean Engineering}}{}
2044 \ifthenelse{\equal{#1}{PPE}}{
2045   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
2046   \global\def\meta@deptname{Planejamento Energético}
2047   \global\def\foreign@deptname{Energy Planning}}{}
2048 \ifthenelse{\equal{#1}{PEP}}{
2049   {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
2050   \global\def\meta@deptname{Engenharia de Produção}
2051   \global\def\foreign@deptname{Production Engineering}}{}
2052 \ifthenelse{\equal{#1}{PEQ}}{
2053   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
2054   \global\def\meta@deptname{Engenharia Química}
2055   \global\def\foreign@deptname{Chemical Engineering}}{}
2056 \ifthenelse{\equal{#1}{PESC}}{
2057   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
2058   \global\def\meta@deptname{Engenharia de Sistemas e Computação}
2059   \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
2060 \ifthenelse{\equal{#1}{PET}}{
2061   {\global\def\local@deptname{Engenharia de Transportes}
2062   \global\def\meta@deptname{Engenharia de Transportes}
2063   \global\def\foreign@deptname{Transportation Engineering}}{}
2064 \ifthenelse{\equal{#1}{PENT}}{
2065   {\global\def\local@deptname{Engenharia de Nanotecnologia}
2066   \global\def\meta@deptname{Engenharia de Nanotecnologia}
2067   \global\def\foreign@deptname{Nanotechnology Engineering}}{}
2068 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

2069 \renewcommand\title[1]{%
2070   \global\def\local@title{#1}%
2071   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
2072 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

2073 \newcommand\foreigntitle[1]{%
2074   \global\def\foreign@title{#1}%
2075   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
2076 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

2077 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
2078 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
2079 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
2080 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
2081 % rosto, subordinated to the title by a colon.
2082 \newcommand\subtitle[1]{%
2083   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
2084 \newcommand\foreignsubtitle[1]{%

```

```

2085 \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
2086 \newcommand\subtitlein[2]{%
2087 \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
2088 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
2089 % Title as the capa and the folha de rosto print it. Annexes A and B show the
2090 % title in caps and the subtitle in ordinary case after a colon, so the case
2091 % change is applied to the title alone.
2092 \newcommand\coppe@titleblock{%
2093 \MakeUppercase{\coppe@selecttitle}%
2094 \@ifundefined{coppe@subtitle@\coppe@mainlang}{: \coppe@selectsubtitle}}
2095 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
2096 % than one, since a single-volume work says nothing about volumes.
2097 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
2098 % institution; the label switches to the plural on the second call, as the
2099 % advisor label already did.
2100 \newcount\@coadvisor\@coadvisor0
2101 \newcommand\coadvisor[5][\%
2102 \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}
2103 \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
2104 \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
2105 \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
2106 \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
2107 \global\advance\@coadvisor by 1
2108 }
2109 \newcommand\local@coadvisorstring{%
2110 \ifnum\@coadvisor>1
2111 \coppemainstring{coadvisors}%
2112 \else
2113 \coppemainstring{coadvisor}%
2114 \fi}
2115 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
2116 \newcommand\foreign@coadvisorstring{%
2117 \ifnum\@coadvisor>1
2118 \coppeforeignstring{coadvisors}%
2119 \else
2120 \coppeforeignstring{coadvisor}%
2121 \fi}
2122 \newcommand\coppe@nvolumes{1}
2123 \newcommand\coppe@thisvolume{1}
2124 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
2125 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
2126 \newcommand\coppe@volumeline{%
2127 \ifnum\coppe@nvolumes>1
2128 \par\vspace*{4mm}%
2129 \coppemainstring{volumename}\space\coppe@thisvolume\space
2130 \coppemainstring{ofname}\space\coppe@nvolumes
2131 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
2132 % which does not end a paragraph, so without it the city ran on into the volume
2133 % line ("Volume 2 de 3 RIO DE JANEIRO").
2134 \par
2135 \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

2136 \newcount\@advisor\@advisor0

```



```

2137 % The optional first argument is the member's institution, required by
2138 % 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
2139 \newcommand\advisor[5][]{%
2140   \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
2141   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
2142   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
2143   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}
2144   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
2145   \global\advance\@advisor by 1
2146   \ifnum\@advisor>1
2147     \renewcommand\local@advisorstring{Orientadores}
2148     \renewcommand\foreign@advisorstring{Advisors}
2149   \fi
2150 }

```

\examiner

```

2151 \newcount\@examiner\@examiner0
2152 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2153 % titulacao every document passed was silently thrown away and never printed,
2154 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
2155 % optional first argument carries the institution.
2156 % CoppeExaminer: is retained with its old meaning for anything reading it.
2157 \newcommand\examiner[4][]{%
2158   \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
2159   \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
2160   \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
2161   \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
2162   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
2163   \global\advance\@examiner by 1
2164 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

2165 \renewcommand\author[2]{%
2166   \global\def\@authname{#1}
2167   \global\def\@authsurn{#2}
2168 }

```

\date Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (\coppe@docmonth / \coppe@docyear) rather than mutating TeX's \month / \year primitives, which would clobber \today.

```

2169 \renewcommand\date[2]{%
2170   \def\coppe@docmonth{#1}%
2171   \def\coppe@docyear{#2}}

```

\local@monthname

```

2172 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2173   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
2174   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

\foreign@monthname

```

2175 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2176   January\or February\or March\or April\or May\or June\or
2177   July\or August\or September\or October\or November\or December\fi}

```

\keyword

```

2178 \newcounter{keywords}
2179 \newcommand\keyword[1]{%
2180   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
2181   \global\addtocounter{keywords}{1}
2182 }
2183 % Keywords for the foreign abstract. Annex F of the manual shows the English
2184 % abstract closing with English keywords, and \keyword alone cannot serve both
2185 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2186 % the \keyword list, which is what a document written before v4.1 expects.
2187 \newcounter{foreignkeywords}
2188 \newcommand\foreignkeyword[1]{%
2189   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2190   \global\addtocounter{foreignkeywords}{1}
2191 }
2192 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2193 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2194 % would close with the main-language keywords under a Portuguese label, which
2195 % is what a Spanish-, French- or Italian-main work would produce.
2196 \newcounter{braziliankeywords}
2197 \newcommand\braziliankeyword[1]{%
2198   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2199   \global\addtocounter{braziliankeywords}{1}
2200 }
2201 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2202 % separated from one another by semicolons.
2203 % #1 label, #2 csname prefix, #3 how many.
2204 \newcommand\coppe@printkeywords[3]{%
2205   \ifnum#3>0
2206     \par\vspace*{7mm}%
2207     \noindent{\singlespacing\textbf{#1:}}\enspace
2208     \count1=0
2209     \@whilenum\count1<#3\do{%
2210       \ifnum\count1>0 ;\space\fi
2211       \csname #2\the\count1\endcsname
2212       \advance\count1 by 1}%
2213     .\par}%
2214   \fi}

```

\areaconcentracao Data for the folha adicional (Coleta CAPES) and for the natureza block of the
 \linhapesquisa folha de rosto and folha de aprovacao. Every one of them defaults to empty;
 \tipoproducao \makefolhaadicional prints an empty rule where a value is missing, so the pro-
 \projetovinculado gramme can complete the sheet by hand if need be.

```

\nomeprojeto 2215 \newcommand\coppe@areaconc{}
\agenciafomento 2216 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
2217 \newcommand\coppe@dataaprov{}
2218 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
2219 \newcommand\coppe@linhapesq{}
2220 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
2221 \newcommand\coppe@tipoproducao{}
2222 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
2223 \newcommand\coppe@projetovinc{}
2224 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
2225 \newcommand\coppe@nomeprojeto{}

```

```

2226 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
2227 \newcounter{coppeagencias}
2228 \newcommand\agenciafomento[2]{%
2229   \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
2230   \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
2231   \global\addtocounter{coppeagencias}{1}}

```

`\freeconfig` This command allows easy changing of core class parameters.

```

2232 \newcommand\freeconfig[6]{%
2233   \providecommand\@degree{}
2234   \renewcommand\@degree{#1}
2235   \providecommand\@degreename{}
2236   \renewcommand\@degreename{#2}
2237   \providecommand\local@degname{}
2238   \renewcommand\local@degname{#3}
2239   \providecommand\foreign@degname{}
2240   \renewcommand\foreign@degname{#4}
2241   \providecommand\local@doctype{}
2242   \renewcommand\local@doctype{#5}
2243   \providecommand\foreign@doctype{}
2244   \renewcommand\foreign@doctype{#6}%
2245 }
2246 % \end{macrocode}
2247 % \end{macro}
2248 %
2249 % \begin{macro}{\frontmatter}
2250 % The number of pages for both frontmatter and mainmatter printed
2251 % in the cataloging details page is computed by means of simple
2252 % \LaTeX\ labels.
2253 % \begin{macrocode}
2254 \renewcommand\frontmatter{%
2255   \cleardoublepage
2256   \@mainmatterfalse
2257 % The count already started at the folha de rosto (see \maketitle) and simply
2258 % continues here: 2.7 requires every folha from the folha de rosto onwards to
2259 % be counted sequentially, printing arabic numerals only from the first textual
2260 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2261 % Introducao come out as page 1, which the norm does not allow.
2262 %
2263 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2264 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2265 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2266 % option warns.
2267 \if@copperoman
2268   \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2269     on the pre-textual folhas and restarts the arabic count at 1 in the
2270     textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2271     those folhas counted but not numbered, and the arabic numbering
2272     continuing the count. Drop the option to comply}%
2273   \pagenumbering{roman}%
2274 \fi
2275 % \thispagestyle vale por UMA pagina. Se a folha de aprovacao transbordar --
2276 % uma banca grande com titulo longo consegue --, a pagina seguinte pegava o
2277 % estilo corrente e saia com folio, na parte pre-textual, onde a 2.7 nao

```

```

2278 % admite. 0 grupo com \pagestyle{empty} cobre quantas paginas forem, e o
2279 % \endgroup devolve o estilo anterior.
2280 \begingroup
2281   \pagestyle{empty}\thispagestyle{empty}
2282   \singlespacing
2283   \makefrontpage
2284   \clearpage
2285 \endgroup
2286 \if@copperoman
2287   \fancypagestyle{plain}{\fancyhf{}}\fancyhead[R0,LE]{\coppe@folio}%
2288   \renewcommand{\headrulewidth}{0pt}}%
2289   \pagestyle{coppe}%
2290 \else
2291   \fancypagestyle{plain}{\fancyhf{}}\renewcommand{\headrulewidth}{0pt}}%
2292   \pagestyle{empty}%
2293 \fi
2294 }

```

\mainmatter

```

2295 \renewcommand\mainmatter{%
2296   \coppe@mainBegin
2297   \cleardoublepage
2298   \@mainmattertrue
2299   \fancypagestyle{plain}{\fancyhf{}}\fancyhead[R0,LE]{\coppe@folio}%
2300   \renewcommand{\headrulewidth}{0pt}}%
2301   \pagestyle{coppe}%
2302 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2303 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2304 % arabic and running since the folha de rosto; all that changes here is that
2305 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2306 % switched to roman in the front matter, has to switch back -- and there the
2307 % restart at 1 is intrinsic to that (non-conforming) scheme.
2308 \if@copperoman\pagenumbering{arabic}\fi

```

\backmatter

```

2309 \renewcommand\backmatter{%
2310   \if@openright
2311     \cleardoublepage
2312   \else
2313     \clearpage
2314   \fi}
2315 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with

```

\renewcommand if the document uses a non-default font or follows an earlier edition
of the norm.
2316 \providecommand{\coppefinalengine}{%
2317   \ifPDFTeX\hologo{pdfLaTeX}\fi
2318   \ifLuaTeX\hologo{LuaLaTeX}\fi
2319   \ifXeTeX\hologo{XeLaTeX}\fi}
2320 % A classe carrega lmodern (as bitmap da Computer Modern saiam em Type 3 e
2321 % inviabilizavam o PDF/A), entao o colofao diz Latin Modern -- quem trocar a
2322 % fonte redefine este comando.
2323 \providecommand{\coppefinalfont}{%
2324   \if@coppesansserif
2325     Latin Modern Sans (\texttt{lmodern})%
2326   \else
2327     Latin Modern (\texttt{lmodern})%
2328   \fi}
2329 \providecommand{\coppefinalmanual}{%
2330   \emph{Norma para a Elabora\c c\~ao Gr\'afica de Teses e Disserta\c c\~oes
2331     da COPPE/UFRJ}}
2332 \newcommand{\coppetexfinalpage}{%
2333   \clearpage
2334   % Force the colophon onto a verso (even) page so it faces the rear cover.
2335   % If the current page is odd (recto), fill it with an empty verso first.
2336   \if@twoside
2337     \ifodd\c@page
2338       \hbox{}\thispagestyle{empty}\newpage
2339     \fi
2340   \fi
2341   \thispagestyle{empty}%
2342   \null\vfill
2343   \begin{otherlanguage}{brazilian}%
2344     \begin{center}
2345       {\footnotesize\setstretch{1.0}}%
2346       \textsc{Colof\~ao}\par
2347       \vspace{0.7em}
2348       \begin{minipage}{0.75\textwidth}\centering
2349         Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2350         em \today, na fonte \coppefinalfont. A bibliografia foi processada
2351         por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2352         vers\~ao\coppe@version, dispon\'i vel no reposit\'orio GitHub
2353         \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2354         conformidade com a \coppefinalmanual.
2355       \end{minipage}\par
2356     \end{center}
2357   \end{otherlanguage}%
2358   \vspace*{2cm}%
2359   \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2360 \newcommand\makecover{%
2361   \begin{titlepage}%
2362   % A capa e um modelo institucional, e sai em espaco simples qualquer que seja o
2363   % espacamento do corpo. Sob a opcao doublespacing ela transbordava para uma
2364   % segunda e ate uma terceira folha -- que ainda por cima levavam folio, porque
2365   % o \thispagestyle do titlepage vale por uma pagina so. 0 \pagestyle{empty} e
2366   % local ao ambiente e cobre quantas paginas forem.
2367   \pagestyle{empty}%
2368   \singlespacing
2369   \centering
2370   \makebox[\linewidth]{%
2371     \includegraphics[height=2cm]{ufrj-logo}\hfill
2372     \includegraphics[height=2cm]{coppe-logo}}\par
2373   \vspace{1.5cm}
2374   {\singlespacing
2375   \MakeUppercase{\local@universityname}\par
2376     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2377       P{\ ' o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia}}\par
2378     \MakeUppercase{Programa de \local@deptname}\par}
2379   \vspace{\stretch{2}}
2380   \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2381   \vspace{\stretch{2}}
2382   \nohyphens{\coppe@titleblock}\par
2383   \coppe@volumeline
2384   \vspace{\stretch{3}}
2385   \MakeUppercase{\local@cityname}\par
2386   \coppe@docyear
2387   \end{titlepage}%
2388 }

```

\maketitle

```

2389 \renewcommand\maketitle{%
2390 % The capa is neither counted nor numbered (2.7), so it lives in its own
2391 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2392 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2393 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2394 % until the textual part starts.
2395 \pagenumbering{alph}
2396 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2397 \makecover
2398 \pagenumbering{arabic}
2399 \begin{titlepage}
2400 % Mesma razao da capa: modelo institucional, espaco simples, e sem folio ainda
2401 % que transborde.
2402 \pagestyle{empty}%
2403 \singlespacing
2404 \begin{flushleft}
2405 \vspace*{1.5mm}
2406 \setlength\baselineskip{0pt}
2407 \setlength\parskip{1mm}
2408 \makebox[\linewidth]{%
2409   \includegraphics[height=2cm]{ufrj-logo}\hfill
2410   \includegraphics[height=2cm]{coppe-logo}}
2411 \end{flushleft}

```

```

2412 \vspace{1.05cm}
2413 \begin{center}
2414 \nohyphens{\@authname\ \@authsurname}\par
2415 \vspace*{3cm}
2416 \nohyphens{\coppe@titleblock}\par
2417 \coppe@volumeline
2418 \end{center}
2419 \vspace*{2.1cm}
2420 \begin{flushright}
2421 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2422 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2423 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2424 % the margins ever change -- the old fixed 8.45cm did not.
2425 \begin{minipage}{0.5\textwidth}
2426 \singlespacing\frontcover@maintext
2427 \end{minipage}\par
2428 \end{flushright}
2429 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2430 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2431 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2432 % right-hand block, which is what a recent review flagged.
2433 \vspace*{12mm}
2434 {\singlespacing
2435 \ifthenelse{\equal{\coppe@linhapesq}{}}{\%
2436 \noindent Linha de pesquisa: \coppe@linhapesq\par
2437 \vspace*{5mm}}{\%
2438 \nohyphens{\%
2439 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2440 \local@advisorstring: &
2441 \count1=0
2442 \toks@={}
2443 \@whilenum \count1<\@advisor \do{\%
2444 \ifcase\count1 % same as \ifnum0=\count1
2445 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2446 \expandafter\endcsname\expandafter\space%
2447 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2448 \else
2449 \toks@=\expandafter\expandafter\expandafter{\%
2450 \expandafter\the\expandafter\toks@%
2451 \expandafter&\expandafter\space%
2452 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2453 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2454 }%
2455 \fi
2456 \advance\count1 by 1}
2457 \the\toks@
2458 \end{tabularx}}}%
2459 \ifnum\@coadvisor>0
2460 \par\vspace*{2mm}%
2461 \nohyphens{\%
2462 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2463 \local@coadvisorstring: &
2464 \count1=0
2465 \toks@={}

```

```

2466 \@whilenum \count1<\@coadvisor \do{%
2467 \ifcase\count1
2468 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2469 \expandafter\endcsname\expandafter\space%
2470 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2471 \else
2472 \toks@=\expandafter\expandafter\expandafter{%
2473 \expandafter\the\expandafter\toks@%
2474 \expandafter&\expandafter\space%
2475 \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2476 \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2477 }%
2478 \fi
2479 \advance\count1 by 1}
2480 \the\toks@
2481 \end{tabularx}}}%
2482 \fi
2483 \par}
2484 \vspace*{\fill}
2485 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2486 % the revised final version is handed to the Programa. The month was never asked
2487 % for here (the capa, Annex A, also shows only the year), and printing it
2488 % invited confusion with the defence date. \date still takes month and year:
2489 % the month is what the abstract pages and the colophon use.
2490 \begin{center}
2491 \local@cityname\par
2492 \coppe@docyear
2493 \end{center}
2494 \end{titlepage}
2495 % Folha adicional (3.1.2.1.2): a sheet of its own right after the folha de
2496 % rosto, ahead of the approval sheet produced by \frontmatter. Until the 2026
2497 % edition this content lived on the VERSO of the folha de rosto (NBR 14724,
2498 % R57); the verso is gone with the printed copy.
2499 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2500 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2501 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2502 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2503 % whatever follows it is folha 2.
2504 \setcounter{page}{2}%
2505 % The page carrying the ficha catalografica is explicitly excluded from the
2506 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2507 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2508 % the same of the verso of the folha de rosto. Hence the counter is stepped
2509 % back after it.
2510 % An exame de qualificacao is not deposited with the library: no ficha
2511 % catalografica, and no Coleta CAPES sheet. Everything else gets both.
2512 \if@coppeexame\else
2513 \makefolhaadicional\clearpage\addtocounter{page}{-1}%
2514 \fi
2515 \global\let\maketitle\relax%
2516 \global\let\and\relax}

2517 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2518 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de

```



```

2519 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2520 % this page opened with the title in caps and the author under it, and set the
2521 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2522 % One board member. \coppe@membroid is the identification line required by
2523 % 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
2524 % when not given, so a document that never supplied them looks as it did before.
2525 \newcommand\coppe@membroid[4]{%
2526   #1\ #2%
2527   \ifthenelse{\equal{#3}{}}{},{, #3}%
2528   \ifthenelse{\equal{#4}{}}{},{, #4}%
2529 % The gap above each signature rule is \coppe@membrogap, set from the size of
2530 % the board just before the list is assembled (see \makefrontpage). A fixed 9mm
2531 % fitted five members and overflowed at seven -- the sheet silently continued
2532 % onto a second page, which also printed a folio in the pre-textual part, where
2533 % 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
2534 % skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
2535 % real glue, and a title long enough to wrap an extra line is absorbed instead
2536 % of costing a page. It has to go through \vspace and not a bare \vskip: the
2537 % board list is emitted inside \nohyphens, i.e. in horizontal mode, where
2538 % \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
2539 % ordinary text -- which is exactly what landed on the page the first time.
2540 \newlength\coppe@membrogap
2541 \setlength\coppe@membrogap{9mm minus 2mm}
2542 \newcount\coppe@bancasize
2543 \newcommand\coppe@membro[4]{%
2544   \if@coppeassin
2545     \vspace{\coppe@membrogap}%
2546     \begin{center}
2547       \rule{95mm}{0.4pt}\par
2548       \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
2549     \end{center}%
2550   \else
2551     \noindent\hspace*{8mm}%
2552     \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2553       \coppe@membroid{#1}{#2}{#3}{#4}
2554     \end{minipage}\par\vspace{2mm}%
2555   \fi}
2556 \newcommand\makefrontpage{%
2557   \begin{center}
2558     \nohyphens{\@authname\ \@authsurn}\par
2559     \vspace*{7mm}
2560     \sloppy\nohyphens{\coppe@selecttitle
2561       \@ifundefined{coppe@subtitle@\coppe@mainlang}{%
2562         \coppe@selectsubtitle}}\par
2563   \end{center}\par
2564   \vspace*{8mm}
2565 % Same block, same indentation and same single spacing as the folha de rosto:
2566 % 2.4 names the two pages together.
2567   \begin{flushright}
2568     \begin{minipage}{0.5\textwidth}
2569       \singlespacing\frontcover@maintext
2570     \end{minipage}
2571   \end{flushright}
2572   \vspace*{12mm}

```

2573 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2574 % filled in; \dataaprovacao supplies it when the date is already known.

2575 \noindent\local@approvedonname:\enspace
2576 \ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
2577 \vspace*{12mm}

2578 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2579 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2580 % above a second list headed "Aprovada por:", which split the board in two and
2581 % did not mark the advisor as its president. They are now one list.

2582 %

2583 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2584 % being built inside a tabularx: each entry now carries four fields (treatment,
2585 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2586 % could not carry them without becoming unreadable. \protected@edef keeps
2587 % accents and other protected commands in the names intact.

2588 % Size of the board decides the spacing above. The thresholds are measured, not
2589 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2590 % second page, 7mm brought six back onto one sheet, 5mm did the same for seven,
2591 % and eight needed 3.5mm -- eight came up in the adversarial suite, in a
2592 % Spanish thesis whose longer natureza block leaves less room than the
2593 % Portuguese one. Boards of five or fewer keep the original 9mm, so every
2594 % document written before this change renders exactly as it did.

2595 \coppe@bancasize=\@advisor
2596 \advance\coppe@bancasize by \@coadvisor
2597 \advance\coppe@bancasize by \@examiner
2598 \ifnum\coppe@bancasize>7
2599 \setlength\coppe@membrogap{3.5mm minus 1.5mm}%
2600 \else\ifnum\coppe@bancasize>6
2601 \setlength\coppe@membrogap{5mm minus 2mm}%
2602 \else\ifnum\coppe@bancasize>5
2603 \setlength\coppe@membrogap{7mm minus 2mm}%
2604 \fi\fi\fi
2605 \gdef\coppe@bancalist{}%
2606 \count1=0
2607 \@whilenum\count1<\@advisor\do{%
2608 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2609 {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2610 {\csname CoppeAdvisorName:\the\count1\endcsname\space
2611 \csname CoppeAdvisorSurname:\the\count1\endcsname}%
2612 {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2613 {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2614 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2615 \expandafter{\coppe@tmp}%
2616 \advance\count1 by 1}%
2617 % Coorientadores come right after the orientadores and before the rest of the
2618 % board, keeping the order 3.1.2.1.3(e) prescribes.

2619 \count1=0
2620 \@whilenum\count1<\@coadvisor\do{%
2621 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2622 {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2623 {\csname CoppeCoadvisorName:\the\count1\endcsname\space
2624 \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2625 {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2626 {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%

```

2627 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2628 \expandafter{\coppe@tmp}%
2629 \advance\count1 by 1}%
2630 \count1=0
2631 \@whilenum\count1<\@examiner\do{%
2632 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2633 {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2634 {\csname CoppeExaminerName:\the\count1\endcsname}%
2635 {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2636 {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2637 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2638 \expandafter{\coppe@tmp}%
2639 \advance\count1 by 1}%
2640 \noindent\local@approvedname:\par
2641 \nohyphens{\singlespacing\coppe@bancalist}\par
2642 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2643 % examining board, and Annex D shows nothing below the signature lines. The date
2644 % that belongs on this page is the date of APPROVAL, printed above.
2645 % \frontpage@bottomtext stays defined for anyone calling it directly.
2646 \vspace*{\fill}}

2647 \newcommand\coppe@hypersetup{%
2648 \begingroup
2649 % changes to \toks@ and \count@ are kept local;
2650 % it's not necessary for them, but it is usually the case
2651 % for \count1, because the first ten counters are written
2652 % to the DVI file, thus you got lucky because of PDF output
2653 \toks@={}% in this special case not necessary
2654 \count@=0 %
2655 \@whilenum\count@<\value{keywords}\do{%
2656 % * a keyword separator is not necessary,
2657 % if there is just one keyword
2658 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2659 % at least once, to get rid of the loop depended \count@
2660 \ifcase\count@ % same as \ifnum0=\count@
2661 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2662 \else
2663 \toks@=\expandafter\expandafter\expandafter{%
2664 \expandafter\the\expandafter\toks@
2665 \expandafter;\expandafter\space
2666 \csname CoppeKeyword:\the\count@\endcsname
2667 }%
2668 \fi
2669 \advance\count@ by 1 %
2670 }%
2671 \edef\x{\endgroup
2672 \noexpand\hypersetup{%
2673 pdfkeywords={\the\toks@}%
2674 }%
2675 }%
2676 \x
2677 \hypersetup{%
2678 pdfauthor={\@authname\ \@authsurn},
2679 pdftitle={\local@title},

```

```

2680 pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
2681 pdfcreator={LaTeX with CoppeTeX toolkit},
2682 breaklinks={true},
2683 raiselinks={true},
2684 pageanchor={true},
2685 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.,” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\@isundefined` (cf. the T_EX FAQ).

```

2686 \newcommand\makecatalog{%
2687   \vspace*{\fill}
2688   \begin{center}
2689     \setlength{\fboxsep}{5mm}
2690     \framebox[120mm][c]{\makebox[5mm][c]{}%
2691       \begin{minipage}[c]{105mm}
2692         \setlength{\parindent}{5mm}
2693         \noindent\sloppy\nohyphens\@authsurn,
2694         \nohyphens\@authname\par
2695         \nohyphens{%
2696           \coppe@selecttitle/\@authname\ \@authsurn. -- \local@cityname:
2697           UFRJ/COPPE, \coppe@docyear.}\par
2698         \pageref{front:pageno},
2699         \pageref{LastPage}
2700         p.\@ifundefined{r@cat:lofflag}{}{\pageref{cat:lofflag}} $29,7$cm.\par
2701         % There is an issue here. When the last entry must be split between lines,
2702         % the spacing between it and the next paragraph becomes smaller.
2703         % Should we manually introduce a fixed space? But how could we know that
2704         % a name was split? Is this happening yet?
2705         \nohyphens{%
2706           \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2707             \local@advisorstring: &
2708             \count1=0
2709             \toks@={}
2710             \@whilenum \count1<\@advisor \do{%
2711               \ifcase\count1 % same as \ifnum0=\count1
2712                 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2713                   \expandafter\endcsname\expandafter\space%
2714                   \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2715               \else
2716                 \toks@=\expandafter\expandafter\expandafter{%
2717                   \expandafter\the\expandafter\toks@
2718                   \expandafter&\expandafter\space
2719                   \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2720                   \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2721               }%
2722             \fi
2723             \advance\count1 by 1}
2724           \the\toks@
2725         \end{tabularx}}\par
2726         \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2727         UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2728         Refer{\^ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par

```

```

2729 \count1=0
2730 \count2=1
2731 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2732 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2733 \advance\count1 by 1
2734 \advance\count2 by 1}
2735 I. \csname CoppeAdvisorSurname:0\endcsname,%
2736 \ \csname CoppeAdvisorName:0\endcsname%
2737 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2738 II. \local@universityname, COPPE, Programa de \local@deptname.
2739 III. T{\ i}tulo.}
2740 \end{minipage}}
2741 \end{center}
2742 \vspace*{\fill}}

```

\fichacatalografica The folha adicional required from August 2026 (SiBI manual, 9th rev. ed.,
\makefolhaaditional 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha cata-
lografica, sitting immediately after the folha de rosto and counted by no-
body. \fichacatalografica takes the PDF produced by the SiBI generator at
<<http://fichacatalografica.sibi.ufrj.br/>>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main lan-
guage of the work – and deliberately without \selectlanguage, since the brazilian
babel language need not be loaded in, say, a Spanish-main document.

```

2743 \newcommand\coppe@fichafile{}
2744 \newcommand\coppe@fichapage{1}
2745 \newcommand\fichacatalografica[2][1]{%
2746 \gdef\coppe@fichapage{#1}\gdef\coppe@fichafile{#2}}
2747 \newcommand\coppe@marcatipo[1]{%
2748 \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2749 \newcommand\coppe@marcavinc[1]{%
2750 \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2751 \newcommand\coppe@campo[1]{%
2752 \ifthenelse{\equal{#1}{}}{\hrulefill}{#1\enspace\hrulefill}}
2753 \newcommand\coppe@checativo{%
2754 \ifthenelse{\equal{\coppe@tipoproducao}{}}{}}{%
2755 \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}\OR
2756 \equal{\coppe@tipoproducao}{artistica}}\OR
2757 \equal{\coppe@tipoproducao}{tecnologica}}\OR
2758 \equal{\coppe@tipoproducao}{tecnica}}{}}{%
2759 \ClassWarning{coppe}{Unknown \string\tipoproducao{ } value
2760 '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2761 bibliografica, artistica, tecnologica or tecnica; no box will be
2762 ticked}}}}
2763 \newcommand\makefolhaadicional{%
2764 \clearpage
2765 \thispagestyle{empty}%
2766 \coppe@checativo
2767 {\singlespacing
2768 \begin{center}
2769 {\large\bfseries Informa\c c\~oes Coleta CAPES}
2770 \end{center}
2771 \vspace{7mm}
2772 \noindent Tipo de produ\c c\~ao intelectual:\par

```

```

2773 \vspace{1mm}
2774 \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
2775 \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art'\i}stica\par
2776 \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2777 \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
2778 \vspace{5mm}
2779 \noindent Projeto de pesquisa vinculado \'a produco intelectual?\par
2780 \vspace{1mm}
2781 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2782 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace No\par
2783 \vspace{5mm}
2784 \noindent Nome do projeto de pesquisa:\par
2785 \vspace{1mm}
2786 \noindent\coppe@campo{\coppe@nomeprojeto}\par
2787 \vspace{5mm}
2788 \noindent \'Area de concentraco da produco intelectual:\par
2789 \vspace{1mm}
2790 \noindent\coppe@campo{\coppe@areaconc}\par
2791 \vspace{5mm}
2792 \noindent Agncia(s) de fomento financiadora(s) da produco
2793 intelectual:\par
2794 \vspace{1mm}
2795 \ifthenelse{\value{coppeagencias}=0}{%
2796 \noindent\hrulefill\par}{%
2797 \count1=0
2798 \@whilenum\count1<\value{coppeagencias}\do{%
2799 \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2800 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2801 \advance\count1 by 1}}%
2802 \vspace{9mm}
2803 \begin{center}
2804 {\large\bfseries Ficha catalogr\'afica}\par
2805 \vspace{4mm}
2806 \ifthenelse{\equal{\coppe@fichafile}{}}{%
2807 \if@copperascunho
2808 \makecatalog
2809 \else
2810 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2811 Espao reservado \'a ficha catalogr\'afica.\\[1ex]
2812 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2813 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2814 ou solicite-a \'a biblioteca do seu Programa.}}%
2815 \fi}{%
2816 \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2817 \end{center}
2818 \par}}

```

\dedication

```

2819 \newcommand\dedication[1]{
2820 \gdef\@dedic{#1}
2821 \cleardoublepage
2822 \vspace*{\fill}
2823 \begin{flushright}
2824 \begin{minipage}{60mm}

```

```

2825         \raggedleft \it \normalsize \@dedic
2826         \end{minipage}
2827     \end{flushright}}

```

\coppe@coadvisorrows The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

2828 \newcommand\coppe@coadvisorrows[1]{%
2829     \if@coppecoadvresumo
2830         \ifnum\@coadvisor>0
2831             \par\vspace*{2mm}%
2832             \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2833                 #1: &
2834                 \count1=0
2835                 \toks@={}
2836                 \@whilenum \count1<\@coadvisor \do{%
2837                     \ifcase\count1
2838                         \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2839                             \expandafter\endcsname\expandafter\space%
2840                             \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2841                     \else
2842                         \toks@=\expandafter\expandafter\expandafter{%
2843                             \expandafter\the\expandafter\toks@%
2844                             \expandafter&\expandafter\space%
2845                             \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2846                             \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2847                     }%
2848                 \fi
2849                 \advance\count1 by 1}
2850                 \the\toks@
2851             \end{tabularx}%
2852         \fi
2853     \fi}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

2854 \newenvironment{abstract}{%
2855     \cleardoublepage
2856     \thispagestyle{plain}
2857     \abstract@toptext\par
2858     \vspace*{8.6mm}
2859     \begin{center}
2860         \sloppy\nohyphens{\MakeUppercase\local@title}\par
2861         \vspace*{13.2mm}
2862         \@authname\ \@authsurn \par
2863         \vspace*{7mm}
2864         \local@monthname/\coppe@docyear
2865     \end{center}\par
2866     \vspace*{1.5cm}
2867     \noindent%
2868     \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2869         \local@advisorstring: &

```

```

2870 \count1=0
2871 \toks@={}
2872 \@whilenum \count1<\@advisor \do{%
2873 \ifcase\count1 % same as \ifnum0=\count1
2874 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2875 \expandafter\endcsname\expandafter\space%
2876 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2877 \else
2878 \toks@=\expandafter\expandafter\expandafter{%
2879 \expandafter\the\expandafter\toks@
2880 \expandafter&\expandafter\space
2881 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2882 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2883 }%
2884 \fi
2885 \advance\count1 by 1}
2886 \the\toks@
2887 \end{tabularx}%
2888 \coppe@coadvisorrows{\local@coadvisorstring}\par
2889 \vspace*{2mm}
2890 \noindent\local@deptstring: \local@deptname\par
2891 \vspace*{7mm}}}%
2892 \coppe@printkeywords{\coppemainstring{keywordsname}}%
2893 {CoppeKeyword:}{\value{keywords}}}%
2894 \vspace*{\fill}}

```

foreignabstract (env.)

```

2895 \newenvironment{foreignabstract}{%
2896 \cleardoublepage
2897 \thispagestyle{plain}
2898 \begin{otherlanguage}{\coppe@foreignlang}
2899 \foreignabstract@toptext\par
2900 \vspace*{8.6mm}
2901 \begin{center}
2902 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
2903 \vspace*{13.2mm}
2904 \@authname\ \@authsur \par
2905 \vspace*{7mm}
2906 \foreign@monthname/\coppe@docyear
2907 \end{center}\par
2908 \vspace*{1.5cm}
2909 \noindent%
2910 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2911 \foreign@advisorstring: &
2912 \count1=0
2913 \toks@={}
2914 \@whilenum \count1<\@advisor \do{%
2915 \ifcase\count1 % same as \ifnum0=\count1
2916 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2917 \expandafter\endcsname\expandafter\space%
2918 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2919 \else
2920 \toks@=\expandafter\expandafter\expandafter{%
2921 \expandafter\the\expandafter\toks@

```



```

2922         \expandafter&\expandafter\space
2923         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2924         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2925     }%
2926     \fi
2927     \advance\count1 by 1}
2928     \the\toks@
2929 \end{tabularx}%
2930 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
2931 \vspace*{2mm}
2932 \noindent\foreign@deptstring: \foreign@deptname\par
2933 \vspace*{7mm}}{%
2934 \ifnum\value{foreignkeywords}>0
2935     \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2936     {CoppeForeignKeyword:}{\value{foreignkeywords}}%
2937 \else
2938     \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2939     {CoppeKeyword:}{\value{keywords}}%
2940 \fi
2941 \end{otherlanguage}
2942 \vspace*{\fill}
2943 \global\let\@author\@empty
2944 \global\let\@date\@empty
2945 \global\let\foreign@title\@empty
2946 \global\let\foreign@title\relax
2947 \global\let\local@title\@empty
2948 \global\let\local@title\relax
2949 \global\let\author\relax
2950 \global\let\author\relax
2951 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

2952 \newenvironment{brazilianabstract}{%
2953     \cleardoublepage
2954     \thispagestyle{plain}
2955     \begin{otherlanguage}{brazilian}
2956     \abstract@toptext\par
2957     \vspace*{8.6mm}
2958     \begin{center}
2959         \sloppy\nohyphens{\MakeUppercase\local@title}\par
2960         \vspace*{13.2mm}
2961         \@authname\ \@authsurn \par
2962         \vspace*{7mm}
2963         \local@monthname/\coppe@docyear
2964     \end{center}\par
2965     \vspace*{1.5cm}
2966     \noindent%

```

```

2967 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2968 \local@advisorstring: &
2969 \count1=0
2970 \toks@={}
2971 \@whilenum \count1<\@advisor \do{%
2972 \ifcase\count1
2973 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2974 \expandafter\endcsname\expandafter\space%
2975 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2976 \else
2977 \toks@=\expandafter\expandafter\expandafter{%
2978 \expandafter\the\expandafter\toks@
2979 \expandafter&\expandafter\space
2980 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2981 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2982 }%
2983 \fi
2984 \advance\count1 by 1}
2985 \the\toks@
2986 \end{tabularx}%
2987 \coppe@coadvisorrows{\local@coadvisorstring}\par
2988 \vspace*{2mm}
2989 \noindent\local@deptstring: \local@deptname\par
2990 \vspace*{7mm}}}%
2991 % \coppestring takes ONE argument and resolves against the current language;
2992 % this runs before \end{otherlanguage}, so \language is already brazilian.
2993 \ifnum\value{braziliankeywords}>0
2994 \coppe@printkeywords{\coppestring[keywordsname]}%
2995 {CoppeBrazilianKeyword:}\value{braziliankeywords}}%
2996 \else
2997 \coppe@printkeywords{\coppestring[keywordsname]}%
2998 {CoppeKeyword:}\value{keywords}}%
2999 \fi
3000 \end{otherlanguage}
3001 \vspace*{\fill}}

```

\listoffigures

```

3002 \renewcommand\listoffigures{%
3003 \coppe@hasLof
3004 \if@twocolumn
3005 \@restonecoltrue\onecolumn
3006 \else
3007 \@restonecolfalse
3008 \fi
3009 \chapter*{\listfigurename}%
3010 \coppe@listtoc{\listfigurename}%
3011 \@mkboth{\MakeUppercase\listfigurename}%
3012 {\MakeUppercase\listfigurename}%
3013 \@starttoc{lof}%
3014 \if@restonecol\twocolumn\fi
3015 }

```

\listoftables

```

3016 \renewcommand\listoftables{%

```

```

3017 \if@twocolumn
3018 \@restonecoltrue\onecolumn
3019 \else
3020 \@restonecolfalse
3021 \fi
3022 \chapter*{\listtablename}%
3023 \coppe@listtoc{\listtablename}%
3024 \@mkboth{%
3025 \MakeUppercase\listtablename}%
3026 {\MakeUppercase\listtablename}%
3027 \@starttoc{lot}%
3028 \if@restonecol\twocolumn\fi
3029 }

```

\printlosymbols

```

3030 \newcommand\printlosymbols{%
3031 \renewcommand\glossaryname{\listsymbolname}%
3032 \begingroup\@coppeglosspretrue
3033 \@input@{\jobname.los}\endgroup}

```

\makelosymbols

```

3034 \def\makelosymbols{%
3035 \newwrite\@losfile
3036 \immediate\openout\@losfile=\jobname.syx
3037 \newcommand\symb1[3][\@bsphack\begingroup
3038 \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
3039 \@sanitize%
3040 \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
3041 {Writing index of symbols file \jobname.syx}%
3042 \let\makelosymbols\@empty%
3043 }%
3044 \@onlypreamble\makelosymbols

3045 \AtBeginDocument{%
3046 \ifpackageloaded{hyperref}{%
3047 \newcommand\@wrlos[3]{%
3048 \protected@write\@losfile{%
3049 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
3050 \endgroup%
3051 \@esphack}}{%
3052 \newcommand\@wrlos[3]{%
3053 \protected@write\@losfile{%
3054 {\string\indexentry{#1[#2] #3}{\thepage}}%
3055 \endgroup%
3056 \@esphack}}}%

```

\printloabbreviations

```

3057 \newcommand\printloabbreviations{%
3058 \renewcommand\glossaryname{\listabbreviationname}%
3059 \begingroup\@coppeglosspretrue
3060 \@input@{\jobname.lab}\endgroup}

```

\makeloabbreviations

```

3061 \def\makeloabbreviations{%

```

```

3062 \newwrite\@labfile
3063 \immediate\openout\@labfile=\jobname.abx
3064 \newcommand\abbrev[3][\@bsphack\beginingroup
3065     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
3066     \@sanitize
3067     \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
3068 {Writing index of abbreviations file \jobname.abx}%
3069 \let\makeloabbreviations\@empty
3070 }
3071 \@onlypreamble\makeloabbreviations

```

```

3072 \AtBeginDocument{%
3073 \ifpackageloaded{hyperref}{%
3074 \newcommand\@wrlab[3]{%
3075     \protected@write\@labfile{%
3076         {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
3077     \endgroup%
3078     \@esphack}}{%
3079 \newcommand\@wrlab[3]{%
3080     \protected@write\@labfile{%
3081         {\string\indexentry{#1[#2] #3}{\thepage}}%
3082     \endgroup%
3083     \@esphack}}}%

```

`\newsigla` Acronyms (siglas), manual2024 2.8 / R43. `\newsigla{key}{SIGLA}{full form}`
`\sigla` declares one; `\sigla{key}` prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through `\abbrev`, so `\makeloabbreviations` must be active), then just “SIGLA” on later uses. `\sigla*{key}` forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

3084 \newcommand\newsigla[3]{%
3085 \@namedef{coppe@sigla@s@#1}{#2}%
3086 \@namedef{coppe@sigla@l@#1}{#3}%
3087 \newcommand\coppe@sigla@reg[1]{%
3088 \ifundefined{abbrev}}{%
3089     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
3090 \newcommand\coppe@sigla@full[1]{%
3091 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
3092 \newcommand\coppe@sigla@undef[1]{%
3093 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
3094 \textbf{???#1???}}
3095 \newcommand\coppe@sigla@first[1]{%
3096 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
3097 \newcommand\coppe@sigla@norm[1]{%
3098 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3099     {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
3100     {\@nameuse{coppe@sigla@s@#1}}}%
3101 \newcommand\coppe@sigla@star[1]{%
3102 \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3103     {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
3104 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

3105 %%% \AtBeginDocument{%
3106 %%% \ifpackageloaded{hyperref}{%
3107 %%% \def\@wrlab#1#2{%

```

```

3108 %%%% \protected@write\@labfile{}%
3109 %%%%      {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
3110 %%%%      \endgroup
3111 %%%%      \@esphack}}{%
3112 %%%%      \def\@wrlab#1#2{%
3113 %%%%      \protected@write\@labfile{}%
3114 %%%%      {\string\indexentry{[#1] #2}{\arabic{page}}}%
3115 %%%%      \endgroup
3116 %%%%      \@esphack}}}

3117 % Catalog-label writers. Two language-related traps must be avoided here:
3118 % 1. The label NAMES contain ':' and '.', which babel makes active in
3119 %    some languages (e.g. Brazilian's "shorthand" set). If the write
3120 %    argument is tokenised while one of those is active, pdfTeX aborts
3121 %    with "<inserted text> }\endwrite" because the shorthand expansion
3122 %    injects unbalanced braces. \detokenize emits its argument as
3123 %    catcode-12 characters regardless of the current activation, so the
3124 %    label NAMES survive whatever language is current at writer time.
3125 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
3126 %    produce small-caps roman numerals via \es@scroman, whose expansion
3127 %    contains an \uppercase group that is unbalanced inside \write. The
3128 %    same crash signature appears. The fix is to bypass LaTeX's
3129 %    \roman/\arabic counter wrappers entirely and use the raw e-TeX
3130 %    primitives \romannumeral and \number directly on the \c@page count
3131 %    register --- both are fully expandable, language-agnostic, and
3132 %    always produce plain ASCII digits or lowercase roman letters.
3133 %    \Roman{page} is left in place because babel-spanish only redefines
3134 %    \@Roman under the (non-default) ucroman option.
3135 \AtBeginDocument{%
3136   \ltx@ifpackageloaded{hyperref}{%
3137     \def\coppe@bibEnd{%
3138       \immediate\write\@auxout{%
3139         \string\newlabel{\detokenize{bib:end}}%
3140         {{{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
3141     \def\coppe@bibBegin{%
3142       \immediate\write\@auxout{%
3143         \string\newlabel{\detokenize{bib:begin}}%
3144         {{{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
3145     \def\coppe@mainBegin{%
3146       \immediate\write\@auxout{%
3147         \string\newlabel{\detokenize{front:pageno}}%
3148         {{{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}}}%
3149     \def\coppe@hasLof{%
3150       \immediate\write\@auxout{%
3151         \string\newlabel{\detokenize{cat:lofflag}}%
3152         {{{\detokenize{:~il. ;}}{\detokenize{page.}\romannumeral\c@page}}}}}%
3153   }{%
3154     \def\coppe@bibEnd{%
3155       \immediate\write\@auxout{%
3156         \string\newlabel{\detokenize{bib:end}}{{{ \number\c@page}}}}}%
3157     \def\coppe@bibBegin{%
3158       \immediate\write\@auxout{%
3159         \string\newlabel{\detokenize{bib:begin}}{{{ \number\c@page}}}}}%
3160     \def\coppe@mainBegin{%
3161       \immediate\write\@auxout{%

```

```

3162     \string\newlabel{\detokenize{front:pageno}}{\{\}\Roman{page}}{\}\}\}%
3163     \def\coppe@hasLof{%
3164         \immediate\write\@auxout{%
3165             \string\newlabel{\detokenize{cat:lofflag}}{\{\}\detokenize{~il.;}}{\}\}\}%
3166     }%
3167 }
3168 % The reference list is now produced by biblatex's \printbibliography (see the
3169 % bibliography setup after babel, above). The former thebibliography
3170 % redefinition and its \bibindent are no longer used and have been removed; the
3171 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
3172 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

3173 \newlength{\recuolongquote}
3174 \setlength{\recuolongquote}{4cm}
3175
3176 \newenvironment*{longquote}[1][default]{%
3177     \begin{list}{\}%
3178     \setlength{\leftmargin}{\recuolongquote}%
3179     \setlength{\rightmargin}{0pt}%
3180     \setlength{\itemindent}{0pt}%
3181     \setlength{\listparindent}{0pt}%
3182     \setlength{\parsep}{0pt}%
3183     \setlength{\topsep}{\baselineskip}%
3184 }{%
3185     \item\relax
3186     \footnotesize
3187     \singlespacing
3188     \ifthenelse{\not\equal{#1}{default}}{%
3189         {\itshape\selectlanguage{#1}}%
3190     }{%
3191         \ignorespaces
3192     }{%
3193     \end{list}%
3194     \ignorespacesafterend
3195 }

3196 \newenvironment{theglossary}{%
3197     \if@twocolumn%
3198         \@restonecoltrue\onecolumn%
3199     \else%
3200         \@restonecolfalse%
3201     \fi%
3202     \@mkboth{\MakeUppercase\glossaryname}%
3203     {\MakeUppercase\glossaryname}%
3204     \chapter*{\glossaryname}%
3205     \if@coppeglosspre
3206         \coppe@listtoc{\glossaryname}%
3207     \else
3208         \addcontentsline{toc}{chapter}{\glossaryname}%
3209     \fi
3210     \list{}
3211     {\setlength{\listparindent}{0in}%
3212     \setlength{\labelwidth}{1.0in}%

```

```

3213 \setlength{\leftmargin}{1.5in}%
3214 \setlength{\labelsep}{0.5in}%
3215 \setlength{\itemindent}{0in}}%
3216 \sloppy}%
3217 {\if@restonecol\twocolumn\fi%
3218 \endlist}
3219 %
3220 \renewenvironment{theindex}{%
3221 \if@twocolumn
3222 \@restonecolfalse
3223 \else
3224 \@restonecoltrue
3225 \fi
3226 \twocolumn[\@makeschapterhead{\indexname}]}%
3227 \@mkboth{\MakeUppercase\indexname}%
3228 {\MakeUppercase\indexname}%
3229 \thispagestyle{plain}\parindent\z@
3230 \addcontentsline{toc}{chapter}{\indexname}
3231 \parskip\z@ \@plus .3\p@\relax
3232 \columnseprule \z@
3233 \columnsep 35\p@
3234 \let\item\@idxitem}
3235 {\if@restonecol\onecolumn\else\clearpage\fi}
3236 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3237 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3238 \newcommand\glossaryname{\coppemainstring{glossary}}
3239 %
3240 \newcommand\local@advisorstring{Orientador}
3241 \newcommand\foreign@advisorstring{Advisor}
3242 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3243 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3244 % already set by the time this is used.
3245 \ifthenelse{\boolean{maledoc}}{%
3246 \newcommand\local@approvedname{\coppemainstring{approvedmale}}}%
3247 }{%
3248 \newcommand\local@approvedname{\coppemainstring{approvedfemale}}}%
3249 }
3250 \ifthenelse{\boolean{maledoc}}{%
3251 \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}}%
3252 }{%
3253 \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}}%
3254 }
3255 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
3256 \newcommand\local@deptstring{Programa}
3257 \newcommand\foreign@deptstring{Department}
3258 \newcommand\local@cityname{Rio de Janeiro}
3259 \newcommand\local@statename{RJ}
3260 \newcommand\local@countryname{Brasil}
3261 %
3262 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3263 % name of the institution AND the area de concentracao in this block. The area
3264 % was missing; it is appended as its own sentence when \areaconcentracao was
3265 % given, and simply omitted otherwise.
3266 \newcommand\frontcover@maintext{

```

```

3267 \sloppy\nohyphens{\local@doctype\ de \@degreename\
3268 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3269 ao Programa de P{\` o}s-gradua{\c c}{\~ a}o em \local@deptname,
3270 COPPE, da \local@universityname, como parte dos requisitos
3271 necess{\` a}rios {\` a} obten{\c c}{\~ a}o do t{\` i}tulo de
3272 \local@degname\ em \local@deptname.}%
3273 \ifthenelse{\equal{\coppe@areaconc}{}}{}{}{
3274 \par\vspace{1ex}%
3275 \sloppy\nohyphens{\` Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
3276 }
3277 %
3278 \newcommand\frontpage@maintext{
3279 \noindent {\MakeUppercase\local@doctype}
3280 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3281 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3282 DE P{\` O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
3283 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3284 NECESS{\` A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3285 {\MakeUppercase\local@degname} EM CI{\^ E}NCIAS EM
3286 {\MakeUppercase\local@deptname.\par}}%
3287 }
3288 %
3289 \newcommand\frontpage@bottomtext{%
3290 \begin{center}
3291 {\MakeUppercase{\local@cityname, \local@statename -- \local@countryname}}\par
3292 {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3293 \end{center}}%
3294 }
3295 %
3296 \newcommand\abstract@toptext{%
3297 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3298 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3299 \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
3300 necess{\` a}rios para a obten{\c c}{\~ a}o do grau de
3301 \local@degname\ em Ci{\^ e}ncias (\@degree)}
3302 }
3303 \newcommand\foreignabstract@toptext{%
3304 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3305 COPPE/UFRJ as a partial fulfillment of the requirements for the
3306 degree of \foreign@degname\ of Science (\@degree)}
3307 }
3308 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, `list-of-*` commands, ...). Loading it as late as

possible is the standard rule of thumb so it sees the final version of those macros, including the ones the `coppe` class redefines (e.g. `\@chapter`, `\listofquadros`).

- `biblatex` – must come after `hyperref` so its citation links and back-references hook into `hyperref` correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

3309 \RequirePackage{setspace}
3310 % Apply the default line spacing now that setspace is available. The flag was
3311 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3312 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
3313 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3314 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3315 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3316 % tcolorbox and several graphics packages produce, so a-1b would fail on
3317 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3318 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3319 \if@coppepdfa
3320   \RequirePackage[a-2b]{pdfx}
3321 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3322 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3323 % no extension schema declares it, so it is a DocInfo entry with no XMP
3324 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3325 % that costs nothing to remove. -1 sets every suppression bit.
3326 %
3327 % The two engines spell this differently, and not just in the name. pdfTeX has
3328 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3329 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3330 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3331 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3332 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3333 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3334 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3335 % LuaTeX run: /ID survives.
3336 %
3337 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3338 % variable name makes LuaTeX typeset the name, and the document stops with
3339 % "Missing \begin{document}" -- which is how the first attempt at this was
3340 % caught.
3341 \ifdefined\pdfsuppressptexinfo
3342   \pdfsuppressptexinfo=-1
3343 \else
3344   \ifdefined\pdfvariable
3345     \pdfvariable suppressoptionalinfo=15
3346   \fi
3347 \fi
3348 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3349 % LOADED -- before the preamble has run, so before \title, \author and
3350 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3351 % the NEXT run. The build already runs pdflatex three times for cross-
3352 % references, so the deposited PDF always carries current metadata; only a
3353 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3354 % warning and lacks dc:title.
3355 %

```

```

3356 % The write happens at the END of the document, not at \begin{document}. This
3357 % class has always documented -- and every example follows it -- that \title,
3358 % \author, \department and \keyword go AFTER \begin{document}, next to
3359 % \maketitle. A write at \begin{document} therefore ran before any of them
3360 % existed and died with "Undefined control sequence" on \@authname, so the
3361 % pdfa option was unusable in exactly the style the manual teaches. It only
3362 % ever worked for a document that set its metadata in the preamble. At the end
3363 % everything is set whichever style the author chose, and \immediate still puts
3364 % the file on disk in time for the next run.
3365 %
3366 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3367 % of our own. TeX has sixteen output streams and example.tex was already using
3368 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3369 % \newwrite here failed with "No room for a new \write" on precisely the
3370 % document that most needs validating. enddocument/afteraux fires after LaTeX
3371 % has closed the .aux, which frees that stream for the one \immediate\openout
3372 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3373 % such hook; there the class falls back to \AtEndDocument with a stream of its
3374 % own, which is the pre-existing behaviour.
3375 %
3376 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3377 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3378 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3379 % into the file instead of the title. The .xmpdata then carried the literal
3380 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3381 % document stopped compiling at all -- a single bad run bricked the file until
3382 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3383 % omitted, which pdfx handles, instead of poisoning the next run.
3384 %
3385 % The writes must be \immediate. \protected@write defers to the shipout, and
3386 % the stream is closed here and now, so deferred writes would land nowhere --
3387 % which is exactly what happened first time round: a 0-byte .xmpdata.
3388 % \let\protect\string inside the group gives the protection that
3389 % \protected@write would have given: accents and other robust commands are
3390 % written literally instead of being expanded. pdfx reads the .xmpdata as
3391 % LaTeX, so \a arrives there intact.
3392 \ifdefined\AddToHook
3393   \let\coppe@xmpout\@mainaux
3394   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3395 \else
3396   \newwrite\coppe@xmpout
3397   \let\coppe@xmphook\AtEndDocument
3398 \fi
3399 \coppe@xmphook{%
3400   \gdef\coppe@xmpkeys{%
3401     \count@=0
3402     \@whilenum\count@<\value{keywords}\do{%
3403       \ifnum\count@>0
3404         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3405       \fi
3406       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3407       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3408       \expandafter{\coppe@tmp}%
3409       \advance\count@ by 1}%

```

```

3410 \immediate\openout\coppe@xmpout=\jobname.xmpdata
3411 \begingroup
3412 \let\protect\string
3413 \@ifundefined{coppe@title@coppe@mainlang}{}{%
3414 \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3415 \@ifundefined{coppe@authname}{}{%
3416 \immediate\write\coppe@xmpout{%
3417 \string\Author{\coppe@authname\space\coppe@authsur}}}%
3418 \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3419 \ifthenelse{equal{\coppe@deptcode}{}}{}{%
3420 \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3421 \@degreeenname. Programa de \meta@deptname\space(\coppe@deptcode),
3422 COPPE/UFRJ}}}%
3423 \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}%
3424 \@ifundefined{coppe@docyear}{}{%
3425 \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3426 \endgroup
3427 \immediate\closeout\coppe@xmpout}%
3428 \fi
3429 \RequirePackage{hyperref}
3430 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3431 % former natbib+bibtex setup. The numbers class option selects the numeric
3432 % citation style; otherwise the author-date default is used. natbib=true keeps
3433 % \citel/\citep available for users migrating from natbib.
3434 \if@coppenumeric
3435 \RequirePackage[backend=biber,datamodel=coppe,%
3436 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{bibratex}
3437 \else
3438 \RequirePackage[backend=biber,datamodel=coppe,%
3439 bibstyle=coppe,citestyle=coppe,natbib=true]{bibratex}
3440 \fi
3441 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3442 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3443 % "Bibliografia" (babel's default \bibname). The title is rendered through
3444 % \coppestring so it follows the current main language in BOTH the chapter
3445 % heading and the .toc entry (bibratex's \bibstring proved unreliable across
3446 % the .toc write, and babel's \bibname caption override was equally fragile).
3447 \defbibheading{bibliography}{%
3448 \chapter*{\coppestring{references}}%
3449 \addcontentsline{toc}{chapter}{\coppestring{references}}}
3450 % The catalog page range (bib:begin/bib:end) was written by the old
3451 % thebibliography redefinition, which bibratex's \printbibliography does not
3452 % use. bibratex offers \AtBeginBibliography but no end counterpart, so the
3453 % begin label uses that hook and the end label is emitted by wrapping
3454 % \printbibliography to write it just after the reference list.
3455 \AtBeginBibliography{\coppe@bibBegin}
3456 \let\coppe@orig@printbibliography\printbibliography
3457 \renewcommand\printbibliography[1][]{%
3458 \coppe@orig@printbibliography[#1]}%
3459 \coppe@bibEnd}
3460 \endclass
3461 \glossary
3462 actual '='

```

```

3463 quote '!'
3464 level '>'
3465 %%%% delim_0 " , p. "
3466 delim_0 "\\dotfill "
3467 lethead_flag 0
3468 headings_flag 0
3469 preamble
3470 "\n\\begin{theglossary}\n \\makeatletter"
3471 postamble
3472 "\n \\end{theglossary}\n"
3473 </glossary>

3474 <*class>
3475 \newcommand{\annex}{
3476     \renewcommand{\appendixname}{\coppestring{annex}}%
3477     \appendix
3478     \@coppeannextrue
3479 % \appendix has already re-issued the centred chapter format, but it baked in
3480 % the \appendixname of that moment. Re-issue it now that the name is "Anexo".
3481     \coppe@appendixheadings
3482 }
3483 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3484 <*dbx>
3485 \ProvidesFile{coppe.dbx}[2026/09/09 v4.1 CoppeTeX ABNT datamodel]
3486 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3487 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3488 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3489 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3490 % performance) are already in biblatex's standard datamodel.
3491 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3492 % Custom fields.
3493 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "... em <course>)"
3494 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3495 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3496 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3497 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3498 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3499 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3500 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
3501 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
3502 \DeclareDatamodelEntryfields{course,coppedegree}
3503 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3504 {artist,platform,director,producer,version}
3505 \DeclareDatamodelEntryfields[map]{scale}
3506 \DeclareDatamodelEntryfields[article]{caderno}
3507 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3508 </dbx>
3509 <*bbx>
3510 \ProvidesFile{coppe.bbx}[2026/09/09 v4.1 CoppeTeX ABNT bibliography style]
3511 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in

```

```

3512 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3513 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3514 % so load setspace here as well (idempotent).
3515 \RequirePackage{setspace}
3516 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3517 % and numeric modes, so we derive from the standard style, not authoryear.
3518 \RequireBibliographyStyle{standard}
3519
3520 % Note: standard style never dashes repeated authors, which already matches the
3521 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3522 \ExecuteBibliographyOptions{%
3523   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3524   giveninits=false,
3525   uniquename=false,
3526   uniquelist=false,
3527   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3528   sorting=nyt,
3529 }
3530
3531 % Language strings (Disponível em / Acesso em / In, etc.)
3532 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3533 \DeclareLanguageMapping{english}{english-coppe}
3534
3535 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3536 \DeclareNameAlias{author}{family-given}
3537 \DeclareNameAlias{editor}{family-given}
3538 \DeclareNameAlias{translator}{family-given}
3539 \DeclareNameAlias{default}{family-given}
3540 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3541 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3542
3543 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3544 % organizer prints the kind of responsibility in parentheses after the
3545 % name, abbreviated, lowercase and in the singular even for several
3546 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3547 % standard style prints ", Org." (comma, no parentheses) when the editor
3548 % takes the author slot, so override editor+others to wrap the role string
3549 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
3550 % the role via coppe:bookeditor below. The role word itself is the
3551 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3552 % packs (brazilian-coppe.lbx et al.).
3553 \renewbibmacro*{editor+others}{%
3554   \ifbool{expr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3555     {\printnames{editor}%
3556       \setunit*\addspace}%
3557     \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3558     \clearname{editor}}
3559   {}
3560 % Same parenthesised role for drivers that take the editor through the plain
3561 % editor macro instead of editor+others (e.g. @manual).
3562 \renewbibmacro*{editor}{%
3563   \ifbool{expr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3564     {\printnames{editor}%
3565       \setunit*\addspace}%

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```

3566 \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3567 \clearname{editor}}
3568 {}
3569
3570 % Uppercase the surname (family + prefix) only inside the bibliography list;
3571 % citations keep normal caps (post-2020 rule).
3572 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3573 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3574 % field names; the month stays abbreviated via \mkbibmonth.
3575 \protected\def\coppe@abntdate#1#2#3{%
3576 \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3577 \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3578 \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
3579 \AtBeginBibliography{%
3580 \singlespacing % entries single-spaced internally (2.4)
3581 \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3582 \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
3583 \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3584 \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3585 \let\mkbibdateshort\coppe@abntdate
3586 }
3587
3588 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3589 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3590 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3591 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3592 % article AND the next word are uppercased; otherwise just the first word. The
3593 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3594 % sentinel so no spurious space is left before the following punctuation.
3595 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3596 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3597 \coppeUCifart{#1}
3598 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3599 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3600 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3601 \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3602 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3603 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3604 \protected\def\coppeUCifart#1{%
3605 \begingroup
3606 \lowercase{\def\coppeUC@w{#1}}%
3607 \edef\coppeUC@w{\coppeUC@w}%
3608 \expandafter\in@\expandafter{\coppeUC@w}{a,o,as,os,um,uma,uns,umas,the,an,}%
3609 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3610 \endgroup}
3611 \DeclareFieldFormat{title}{%
3612 \ifboolexpr{ test {\ifnameundef{labelname}} }
3613 {#1}%
3614 {\mkbibbold{#1}}}
3615 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3616 % format receives the RAW title text (the title format above only sees the
3617 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3618 % first word here.
3619 \DeclareFieldFormat{titlecase}{%

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```

3620 \ifboolexpr{ test {\ifnameundef{labelname}} }
3621   {\coppeUCtitle{#1}}%
3622   {#1}}
3623 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{
3624 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3625 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3626 % the title of a whole document, so override those back to bold.
3627 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3628 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3629 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3630 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3631 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3632 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
3633 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
3634
3635 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3636 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3637 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
3638 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3639
3640 % --- Periodical volume / number prefixes ---
3641 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
3642 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
3643
3644 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3645 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3646
3647 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3648 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3649
3650 % --- ABNT article driver: no "In."; Journal, Local, v., n., p., date, DOI/URL.
3651 %   Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3652 %   differ: the newspaper name (journaltitle) and city, then either
3653 %       "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3654 %       "p. <pages>, <date>" (page before date, when none),
3655 %   per NBR 6023 §4.2.3.6. ---
3656 \DeclareBibliographyDriver{article}{%
3657   \usebibmacro{bibindex}%
3658   \usebibmacro{begentry}%
3659   \usebibmacro{author/translator+others}%
3660   \setunit{\labelnamepunct}\newblock
3661   \usebibmacro{title}%
3662   \newunit\newblock
3663   \printfield{journaltitle}%
3664   \setunit{\addcomma\space}\printlist{location}%
3665   \iffieldequalstr{entrysubtype}{newspaper}
3666     {\iffieldundef{caderno}
3667       {\setunit{\addcomma\space}\printfield{pages}%
3668         \setunit{\addcomma\space}\printdate}%
3669       {\setunit{\addcomma\space}\printdate%
3670         \setunit{\adddot\space}\printfield{caderno}%
3671         \setunit{\addcomma\space}\printfield{pages}}}%
3672   {\setunit{\addcomma\space}\printfield{volume}%
3673     \setunit{\addcomma\space}\printfield{number}%

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3674 \setunit{\addcomma\space}\printfield{pages}%
3675 \setunit{\addcomma\space}\printdate}%
3676 \newunit\newblock
3677 \printfield{doi}%
3678 \newunit\newblock
3679 \usebibmacro{url+urldate}%
3680 \usebibmacro{finentry}}
3681
3682 % --- incollection: ABNT puts the host organizer before the booktitle, as
3683 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3684 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3685 \newbibmacro*{coppe:bookeditor}{%
3686 \ifnameundef{editor}
3687 {}
3688 {\printnames{editor}%
3689 \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3690 \clearname{editor}%
3691 \setunit{\adddot\space}}}
3692 \newbibmacro*{coppe:partdriver}{%
3693 \usebibmacro{bibindex}\usebibmacro{begentry}%
3694 \usebibmacro{author/translator+others}%
3695 \setunit{\labelnamepunct}\newblock
3696 \usebibmacro{title}%
3697 \newunit\newblock
3698 \usebibmacro{in:}%
3699 \usebibmacro{coppe:bookeditor}%
3700 \printfield{booktitle}%
3701 \newunit\newblock
3702 \printfield{edition}%
3703 \newunit\newblock
3704 \printlist{location}%
3705 \setunit*{\addcolon\space}\printlist{publisher}%
3706 \setunit*{\addcomma\space}\printdate%
3707 \newunit\newblock
3708 \printfield{pages}%
3709 \setunit{\addcomma\space}\printfield{chapter}%
3710 \newunit\newblock
3711 \usebibmacro{url+urldate}%
3712 \usebibmacro{finentry}}
3713 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3714 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3715
3716 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
3717 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3718
3719 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3720 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
3721 \DeclareBibliographyDriver{thesis}{%
3722 \usebibmacro{bibindex}\usebibmacro{begentry}%
3723 \usebibmacro{author}%
3724 \setunit{\labelnamepunct}\newblock
3725 \usebibmacro{title}%
3726 \newunit\newblock
3727 \printfield{type}\setunit*{\addspace}\printfield{course}%

```



```

3728 \newunit\newblock
3729 \printlist{institution}%
3730 \setunit*{\addcomma\space}\printlist{location}%
3731 \setunit*{\addcomma\space}\printdate%
3732 \newunit\newblock
3733 \printfield{pagetotal}%
3734 \newunit\newblock
3735 \printfield{note}%
3736 \newunit\newblock
3737 \usebibmacro{url+urldate}%
3738 \usebibmacro{finentry}}
3739 % --- Games / audiovisual: render the custom fields (artist, platform,
3740 % director, producer, version) the generic misc driver ignores. ---
3741 \newbibmacro*{coppe:media}{%
3742 \iflistundef{artist}{\setunit{\addperiod\space}%
3743 \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3744 \iffieldundef{platform}{\setunit{\addperiod\space}%
3745 \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3746 \iflistundef{director}{\setunit{\addperiod\space}%
3747 \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
3748 \iflistundef{producer}{\setunit{\addperiod\space}%
3749 \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}}%
3750 \newbibmacro*{coppe:mediadriver}{%
3751 \usebibmacro{bibindex}\usebibmacro{begentry}%
3752 \usebibmacro{author/translator+others}%
3753 \setunit{\labelnamepunct}\newblock
3754 \usebibmacro{title}%
3755 \newunit\newblock
3756 \usebibmacro{coppe:media}%
3757 \newunit\newblock
3758 \printlist{publisher}%
3759 \setunit*{\addcomma\space}\printlist{location}%
3760 \setunit*{\addcomma\space}\printdate%
3761 \newunit\newblock
3762 \printfield{version}%
3763 \newunit\newblock
3764 \printfield{note}%
3765 \newunit\newblock
3766 \usebibmacro{url+urldate}%
3767 \usebibmacro{finentry}}
3768 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
3769 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
3770 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
3771 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
3772 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
3773 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
3774 % imprenta) plus the mandatory scale ("Escala 1:N").
3775 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
3776 \DeclareBibliographyDriver{map}{%
3777 \usebibmacro{bibindex}\usebibmacro{begentry}%
3778 \usebibmacro{author/translator+others}%
3779 \setunit{\labelnamepunct}\newblock
3780 \usebibmacro{title}%
3781 \newunit\newblock

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```

3782 \printlist{location}%
3783 \setunit*{\addcolon\space}\printlist{publisher}%
3784 \setunit*{\addcomma\space}\printdate%
3785 \newunit\newblock
3786 \printfield{scale}%
3787 \newunit\newblock
3788 \usebibmacro{url+urldate}%
3789 \usebibmacro{finentry}}
3790 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
3791 % number. date. The holder (depositante) is printed in normal case, not the
3792 % ALL-CAPS family treatment the bibliography gives the leading author.
3793 \newbibmacro*{coppe:depositor}{%
3794 \ifnameundef{holder}
3795 {}
3796 {\printtext{\bibstring{depositor}\addcolon\space}%
3797 \begingroup
3798 \def\mkbibnamefamily##1{##1}%
3799 \def\mkbibnameprefix##1{##1}%
3800 \printnames{holder}%
3801 \endgroup}}
3802 \DeclareBibliographyDriver{patent}{%
3803 \usebibmacro{bibindex}%
3804 \usebibmacro{begentry}%
3805 \usebibmacro{author/translator+others}%
3806 \setunit{\labelnamepunct}\newblock
3807 \usebibmacro{title}%
3808 \newunit\newblock
3809 \usebibmacro{coppe:depositor}%
3810 \newunit\newblock
3811 \printfield{type}%
3812 \setunit*{\addspace}\printfield{number}%
3813 \newunit\newblock
3814 \printdate%
3815 \newunit\newblock
3816 \usebibmacro{url+urldate}%
3817 \usebibmacro{finentry}}
3818 % Exotic ABNT types that biblatex's standard style leaves without a driver:
3819 % give each a baseline (book-like for published docs: norma/partitura; misc for
3820 % the rest) so they print. Per-type ABNT refinements can follow later.
3821 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
3822 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
3823 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
3824 % the law id + ementa (plain, not bold), then location and date.
3825 \newbibmacro*{coppe:legaldriver}{%
3826 \usebibmacro{bibindex}\usebibmacro{begentry}%
3827 \usebibmacro{author/translator+others}%
3828 \setunit{\labelnamepunct}\newblock
3829 \usebibmacro{title}%
3830 \newunit\newblock
3831 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
3832 \iffieldundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\newblock}
3833 \printlist{location}%
3834 \setunit{\addcomma\space}\printdate%
3835 \newunit\newblock

```

```

3836 \usebibmacro{url+urldate}%
3837 \usebibmacro{finentry}}
3838 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
3839 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
3840 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
3841 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
3842 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
3843 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
3844 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
3845 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
3846 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
3847 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
3848
3849 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
3850 \DeclareSourcemap{%
3851 \maps[datatype=bibtex]{%
3852 % entry-type PT synonyms -> canonical English
3853 \map{step[typesource=livro,typetarget=book]}
3854 \map{step[typesource=artigo,typetarget=article]}
3855 \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
3856 \map{step[typesource=partedelivro,typetarget=incollection]}
3857 \map{step[typesource=emlivro,typetarget=inbook]}
3858 \map{step[typesource=verbete,typetarget=inreference]}
3859 \map{step[typesource=anais,typetarget=proceedings]}
3860 \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
3861 \map{step[typesource=periodico,typetarget=periodical]}
3862 \map{step[typesource=relatorio,typetarget=report]}
3863 \map{step[typesource=relatoriotecnico,typetarget=report]}
3864 \map{step[typesource>manual,typetarget>manual]}
3865 \map{step[typesource=correspondencia,typetarget=letter]}
3866 \map{step[typesource=patente,typetarget=patent]}
3867 \map{step[typesource=diverso,typetarget=misc]}
3868 \map{step[typesource=jogo,typetarget=game]}
3869 \map{step[typesource=jogoeletronico,typetarget=videogame]}
3870 \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
3871 \map{step[typesource=filme,typetarget=movie]}
3872 \map{step[typesource=serietv,typetarget=tvshow]}
3873 \map{step[typesource=programadstv,typetarget=tvshow]}
3874 % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
3875 % except map (declared in coppe.dbx).
3876 \map{step[typesource=legislacao,typetarget=legislation]}
3877 \map{step[typesource=atonormativo,typetarget=legislation]}
3878 \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
3879 \map{step[typesource=documentojuridico,typetarget=legal]}
3880 \map{step[typesource=documentocivil,typetarget=legal]}
3881 \map{step[typesource=norma,typetarget=standard]}
3882 \map{step[typesource=partitura,typetarget=music]}
3883 \map{step[typesource=iconografico,typetarget=image]}
3884 \map{step[typesource=obradearte,typetarget=artwork]}
3885 \map{step[typesource=tridimensional,typetarget=artwork]}
3886 \map{step[typesource=cartografico,typetarget=map]}
3887 \map{step[typesource=mapa,typetarget=map]}
3888 \map{step[typesource=audiovisual,typetarget=video]}
3889 \map{step[typesource=conjuntodedados,typetarget=dataset]}

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3890 \map{step[typesource=fasciculo,typetarget=supperiodical]}
3891 \map{step[typesource=trabalhoapresentado,typetarget=unpublished]}
3892 % thesis family: retype to thesis and preset the (babel-aware) type word
3893 \map{step[typesource=tese,typetarget=thesis]}
3894 \map{step[typesource=dissertacao,typetarget=thesis]}
3895 \map{step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3896 \map{step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3897 \map{step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
3898 \map{step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
3899 \map{step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
3900 \map{step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
3901 \map{step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
3902 \map{step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
3903 % field PT synonyms (unaccented) -> canonical English
3904 \map{step[fieldsource=periodico,fieldtarget=journaltitle]}
3905 \map{step[fieldsource=revista,fieldtarget=journaltitle]}
3906 \map{step[fieldsource=jornal,fieldtarget=journaltitle]}
3907 \map{step[fieldsource=titulo,fieldtarget=title]}
3908 \map{step[fieldsource=subtitulo,fieldtarget=subtitle]}
3909 \map{step[fieldsource=autor,fieldtarget=author]}
3910 \map{step[fieldsource=editora,fieldtarget=publisher]}
3911 \map{step[fieldsource=local,fieldtarget=location]}
3912 \map{step[fieldsource=ano,fieldtarget=year]}
3913 \map{step[fieldsource=data,fieldtarget=date]}
3914 \map{step[fieldsource=edicao,fieldtarget=edition]}
3915 \map{step[fieldsource=paginas,fieldtarget=pages]}
3916 \map{step[fieldsource=numero,fieldtarget=number]}
3917 \map{step[fieldsource=tradutor,fieldtarget=translator]}
3918 \map{step[fieldsource=organizador,fieldtarget=editor]}
3919 \map{step[fieldsource=instituicao,fieldtarget=institution]}
3920 \map{step[fieldsource=tipo,fieldtarget=type]}
3921 \map{step[fieldsource=nota,fieldtarget=note]}
3922 \map{step[fieldsource=dataacesso,fieldtarget=urldate]}
3923 \map{step[fieldsource=curso,fieldtarget=course]}
3924 \map{step[fieldsource=grau,fieldtarget=coppedegree]}
3925 \map{step[fieldsource=artista,fieldtarget=artist]}
3926 \map{step[fieldsource=plataforma,fieldtarget=platform]}
3927 \map{step[fieldsource=diretor,fieldtarget=director]}
3928 \map{step[fieldsource=produtor,fieldtarget=producer]}
3929 \map{step[fieldsource=versao,fieldtarget=version]}
3930 \map{step[fieldsource=escala,fieldtarget=scale]}
3931 }%
3932 }
3933
3934 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
3935 \setlength{\bibhang}{0pt}
3936 \setlength{\bibitemsep}{\baselineskip}
3937
3938 </bbx>
3939 <*cbx>
3940 \ProvidesFile{coppe.cbx}[2026/09/09 v4.1 CoppeTeX ABNT author-date citations]
3941 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
3942 % author-year cites with normal-caps names (post-2020 rule) out of the box.
3943 \RequireCitationStyle{authoryear-comp}

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3944
3945 % --- NBR 10520:2023 author separator -----
3946 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
3947 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
3948 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
3949 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
3950 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
3951 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
3952 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
3953 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
3954
3955 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
3956 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
3957 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
3958 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
3959 % so it is never added to the bibliography.
3960 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3961 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
3962 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3963 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
3964 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
3965 \DeclareRobustCommand{\citetapud}[4][]{%
3966 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
3967 \DeclareRobustCommand{\citepapud}[4][]{%
3968 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
3969 \citeauthor{#4}\addcomma\space\citeyear{#4}%
3970 \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}%
3971 \</cbx>
3972 \<*\bxbr>
3973 \ProvidesFile{brazilian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX brazilian strings]
3974 \InheritBibliographyExtras{brazilian}
3975 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3976 \DeclareBibliographyStrings{%
3977 inherit = {brazilian},
3978 availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
3979 urlseen = {{Acesso em}{Acesso em}},
3980 in = {{In}{In}},
3981 editor = {{organizador}{org\adddot}},
3982 editors = {{organizadores}{org\adddot}},
3983 depositor = {{Depositante}{Depositante}},
3984 coppescale = {{Escala}{Escala}},
3985 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
3986 dscthesi = {{Tese de Doutorado}{Tese de Doutorado}},
3987 phdthesis = {{Tese de Doutorado}{Tese de Doutorado}},
3988 tcc = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
3989 }
3990 \</\bxbr>
3991 \<*\bxen>
3992 \ProvidesFile{english-coppe.lbx}[2026/09/09 v4.1 CoppeTeX english strings]
3993 \InheritBibliographyExtras{english}
3994 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3995 \DeclareBibliographyStrings{%
3996 inherit = {english},
3997 availablefrom = {{Available from}{Available from}},

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3998 urlseen      = {{Access on}{Access on}},
3999 in            = {{In}{In}},
4000 editor        = {{editor}{ed\addot}},
4001 editors       = {{editors}{ed\addot}},
4002 depositor     = {{Depositor}{Depositor}},
4003 coppescale    = {{Scale}{Scale}},
4004 msdiss        = {{M\addot Sc\addot Dissertation}{M\addot Sc\addot Diss\addot}},
4005 dscthesi      = {{D\addot Sc\addot Thesis}{D\addot Sc\addot Thesis}},
4006 phdthesis     = {{Ph\addot D\addot Thesis}{Ph\addot D\addot Thesis}},
4007 tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
4008 }
4009 </lbxen>
4010 <*lbxes>
4011 \ProvidesFile{spanish-coppe.lbx}[2026/09/09 v4.1 CoppeTeX spanish strings]
4012 \InheritBibliographyExtras{spanish}
4013 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
4014 \DeclareBibliographyStrings{%
4015   inherit      = {spanish},
4016   availablefrom = {{Disponible en}{Disponible en}},
4017   urlseen      = {{Consultado el}{Consultado el}},
4018   in           = {{En}{En}},
4019   editor       = {{editor}{ed\addot}},
4020   editors      = {{editores}{ed\addot}},
4021   depositor    = {{Solicitante}{Solicitante}},
4022   coppescale   = {{Escala}{Escala}},
4023   msdiss       = {{Tesis de Maestr'\i a}{Tesis de Maestr'\i a}},
4024   dscthesi     = {{Tesis Doctoral}{Tesis Doctoral}},
4025   phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
4026   tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
4027 }
4028 </lbxes>
4029 <*lbxfr>
4030 \ProvidesFile{french-coppe.lbx}[2026/09/09 v4.1 CoppeTeX french strings]
4031 \InheritBibliographyExtras{french}
4032 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
4033 \DeclareBibliographyStrings{%
4034   inherit      = {french},
4035   availablefrom = {{Disponible \a}{Disponible \a}},
4036   urlseen      = {{Consult\ 'e le}{Consult\ 'e le}},
4037   in           = {{Dans}{Dans}},
4038   editor       = {{\ 'editeur}{\ 'ed\addot}},
4039   editors      = {{\ 'editeurs}{\ 'ed\addot}},
4040   depositor    = {{D\ 'eposant}{D\ 'eposant}},
4041   coppescale   = {{\ 'Echelle}{\ 'Echelle}},
4042   msdiss       = {{M\ 'emoire de Master}{M\ 'emoire de Master}},
4043   dscthesi     = {{Th\ 'ese de Doctorat}{Th\ 'ese de Doctorat}},
4044   phdthesis    = {{Th\ 'ese de Doctorat}{Th\ 'ese de Doctorat}},
4045   tcc          = {{M\ 'emoire de fin d\ 'etudes}{M\ 'emoire de fin d\ 'etudes}},
4046 }
4047 </lbxfr>
4048 <*lbxit>
4049 \ProvidesFile{italian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX italian strings]
4050 \InheritBibliographyExtras{italian}
4051 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}

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4052 \DeclareBibliographyStrings{%
4053   inherit      = {italian},
4054   availablefrom = {{Disponibile da}{Disponibile da}},
4055   urlseen      = {{Consultato il}{Consultato il}},
4056   in           = {{In}{In}},
4057   editor       = {{curatore}{cur\adddot}},
4058   editors      = {{curatori}{cur\adddot}},
4059   depositor    = {{Depositante}{Depositante}},
4060   coppescale   = {{Scala}{Scala}},
4061   msdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
4062   dscthesi     = {{Tesi di Dottorato}{Tesi di Dottorato}},
4063   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
4064   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
4065 }
4066 \ltxit
4067 <*langes>
4068 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
4069 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
4070 %% spanish slot) is active. Populates the dispatcher table for the
4071 %% spanish language and registers the biblalex language mapping.
4072 %% Institutional names (universityname/cityname/statename/countryname)
4073 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
4074 %% rosto are a Brazilian institutional template that does not translate.
4075 \ProvidesFile{coppe-lang-spanish.def}[2026/09/09 v4.1 CoppeTeX spanish class strings]
4076 \copperdefstring{spanish}{appendix}      {Ap\endice}
4077 \copperdefstring{spanish}{annex}         {Anexo}
4078 \copperdefstring{spanish}{frame}         {Cuadro}
4079 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
4080 \copperdefstring{spanish}{source}        {Fuente}
4081 \copperdefstring{spanish}{program}       {Programa}
4082 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
4083 \copperdefstring{spanish}{references}    {Referencias}
4084 \copperdefstring{spanish}{in:}          {en}
4085 \copperdefstring{spanish}{art}           {Arte}
4086 \copperdefstring{spanish}{platform}     {Plataforma}
4087 \copperdefstring{spanish}{directedby}    {Direcci'on}
4088 \copperdefstring{spanish}{producedby}    {Producci'on}
4089 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
4090 \copperdefstring{spanish}{listsymbol}    {Lista de S'imbolos}
4091 \copperdefstring{spanish}{glossary}      {Glosario}
4092 \copperdefstring{spanish}{advisor}       {Director}
4093 \copperdefstring{spanish}{advisors}      {Directores}
4094 \copperdefstring{spanish}{dept}          {Programa}
4095 \copperdefstring{spanish}{approvedmale}  {Aprobado por}
4096 \copperdefstring{spanish}{approvedfemale}{Aprobada por}
4097 \copperdefstring{spanish}{approvedonmale}{Aprobado el}
4098 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
4099 \copperdefstring{spanish}{keywordsname} {Palabras clave}
4100 \copperdefstring{spanish}{volumename}   {Volumen}
4101 \copperdefstring{spanish}{coadvisor}     {Codirector}
4102 \copperdefstring{spanish}{coadvisors}    {Codirectores}
4103 \copperdefstring{spanish}{ofname}        {de}
4104 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
4105 \copperdefstring{spanish}{cityname}      {Rio de Janeiro}

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4106 \copperdefstring{spanish}{statename} {RJ}
4107 \copperdefstring{spanish}{countryname} {Brasil}
4108 \copperdefstring{spanish}{monthname1} {enero}
4109 \copperdefstring{spanish}{monthname2} {febrero}
4110 \copperdefstring{spanish}{monthname3} {marzo}
4111 \copperdefstring{spanish}{monthname4} {abril}
4112 \copperdefstring{spanish}{monthname5} {mayo}
4113 \copperdefstring{spanish}{monthname6} {junio}
4114 \copperdefstring{spanish}{monthname7} {julio}
4115 \copperdefstring{spanish}{monthname8} {agosto}
4116 \copperdefstring{spanish}{monthname9} {septiembre}
4117 \copperdefstring{spanish}{monthname10} {octubre}
4118 \copperdefstring{spanish}{monthname11} {noviembre}
4119 \copperdefstring{spanish}{monthname12} {diciembre}
4120 \copperdefstring{spanish}{algorithm2eopt} {spanish}
4121 \copperdefstring{spanish}{babelname} {spanish}
4122 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
4123 %% language-mapping registration until \begin{document} so biblatex's
4124 %% \DeclareLanguageMapping is available.
4125 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
4126 \end{langes}
4127 \end{*langfr}
4128 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
4129 %% See coppe-lang-spanish.def for the structure. Institutional names
4130 %% remain Portuguese; only document-content strings translate.
4131 \ProvidesFile{coppe-lang-french.def}[2026/09/09 v4.1 CoppeTeX french class strings]
4132 \copperdefstring{french}{appendix} {Annexe}
4133 \copperdefstring{french}{annex} {Annexe}
4134 \copperdefstring{french}{frame} {Encadr\'e}
4135 \copperdefstring{french}{listframe} {Liste des encadr\'es}
4136 \copperdefstring{french}{source} {Source}
4137 \copperdefstring{french}{program} {Programme}
4138 \copperdefstring{french}{listprogram} {Liste des programmes}
4139 \copperdefstring{french}{references} {R\'ef\'erences}
4140 \copperdefstring{french}{in:} {dans}
4141 \copperdefstring{french}{art} {Illustration}
4142 \copperdefstring{french}{platform} {Plateforme}
4143 \copperdefstring{french}{directedby} {R\'ealis\'e par}
4144 \copperdefstring{french}{producedby} {Produit par}
4145 \copperdefstring{french}{listabbreviation} {Liste des abr\'eviations et acronymes}
4146 \copperdefstring{french}{listsymbol} {Liste des symboles}
4147 \copperdefstring{french}{glossary} {Glossaire}
4148 \copperdefstring{french}{advisor} {Directeur}
4149 \copperdefstring{french}{advisors} {Directeurs}
4150 \copperdefstring{french}{dept} {Programme}
4151 \copperdefstring{french}{approvedmale} {Approuv\'e par}
4152 \copperdefstring{french}{approvedfemale} {Approuv\'ee par}
4153 \copperdefstring{french}{approvedonmale} {Approuv\'e le}
4154 \copperdefstring{french}{approvedonfemale} {Approuv\'ee le}
4155 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
4156 \copperdefstring{french}{volumename} {Volume}
4157 \copperdefstring{french}{coadvisor} {Co-directeur}
4158 \copperdefstring{french}{coadvisors} {Co-directeurs}
4159 \copperdefstring{french}{ofname} {de}

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4160 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
4161 \copperdefstring{french}{cityname} {Rio de Janeiro}
4162 \copperdefstring{french}{statename} {RJ}
4163 \copperdefstring{french}{countryname} {Brasil}
4164 \copperdefstring{french}{monthname1} {janvier}
4165 \copperdefstring{french}{monthname2} {f\'evrier}
4166 \copperdefstring{french}{monthname3} {mars}
4167 \copperdefstring{french}{monthname4} {avril}
4168 \copperdefstring{french}{monthname5} {mai}
4169 \copperdefstring{french}{monthname6} {juin}
4170 \copperdefstring{french}{monthname7} {juillet}
4171 \copperdefstring{french}{monthname8} {ao\^ut}
4172 \copperdefstring{french}{monthname9} {septembre}
4173 \copperdefstring{french}{monthname10}{octobre}
4174 \copperdefstring{french}{monthname11}{novembre}
4175 \copperdefstring{french}{monthname12}{d\'ecembre}
4176 \copperdefstring{french}{algorithm2eopt}{french}
4177 \copperdefstring{french}{babelname} {french}
4178 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4179 \</langfr>
4180 \*langit)
4181 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
4182 %% See coppe-lang-spanish.def for the structure. The algorithm2e
4183 %% language option for Italian is spelled "italiano" -- that's the
4184 %% spelling algorithm2e ships, even though babel calls the language
4185 %% "italian".
4186 \ProvidesFile{coppe-lang-italian.def}[2026/09/09 v4.1 CoppeTeX italian class strings]
4187 \copperdefstring{italian}{appendix} {Appendice}
4188 \copperdefstring{italian}{annex} {Allegato}
4189 \copperdefstring{italian}{frame} {Riquadro}
4190 \copperdefstring{italian}{listframe} {Elenco dei riquadri}
4191 \copperdefstring{italian}{source} {Fonte}
4192 \copperdefstring{italian}{program} {Programma}
4193 \copperdefstring{italian}{listprogram} {Elenco dei programmi}
4194 \copperdefstring{italian}{references} {Riferimenti}
4195 \copperdefstring{italian}{in:} {in}
4196 \copperdefstring{italian}{art} {Illustrazione}
4197 \copperdefstring{italian}{platform} {Piattaforma}
4198 \copperdefstring{italian}{directedby} {Regia di}
4199 \copperdefstring{italian}{producedby} {Prodotto da}
4200 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
4201 \copperdefstring{italian}{listsymbol} {Elenco dei simboli}
4202 \copperdefstring{italian}{glossary} {Glossario}
4203 \copperdefstring{italian}{advisor} {Relatore}
4204 \copperdefstring{italian}{advisors} {Relatori}
4205 \copperdefstring{italian}{dept} {Programma}
4206 \copperdefstring{italian}{approvedmale} {Approvato da}
4207 \copperdefstring{italian}{approvedfemale} {Approvata da}
4208 \copperdefstring{italian}{approvedonmale} {Approvato il}
4209 \copperdefstring{italian}{approvedonfemale}{Approvata il}
4210 \copperdefstring{italian}{keywordsname} {Parole chiave}
4211 \copperdefstring{italian}{volumename} {Volume}
4212 \copperdefstring{italian}{coadvisor} {Correlatore}
4213 \copperdefstring{italian}{coadvisors} {Correlatori}

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4214 \copperdefstring{italian}{ofname}      {di}
4215 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
4216 \copperdefstring{italian}{cityname}     {Rio de Janeiro}
4217 \copperdefstring{italian}{statename}    {RJ}
4218 \copperdefstring{italian}{countryname}  {Brasil}
4219 \copperdefstring{italian}{monthname1}   {gennaio}
4220 \copperdefstring{italian}{monthname2}   {febbraio}
4221 \copperdefstring{italian}{monthname3}   {marzo}
4222 \copperdefstring{italian}{monthname4}   {aprile}
4223 \copperdefstring{italian}{monthname5}   {maggio}
4224 \copperdefstring{italian}{monthname6}   {giugno}
4225 \copperdefstring{italian}{monthname7}   {luglio}
4226 \copperdefstring{italian}{monthname8}   {agosto}
4227 \copperdefstring{italian}{monthname9}   {settembre}
4228 \copperdefstring{italian}{monthname10} {ottobre}
4229 \copperdefstring{italian}{monthname11} {novembre}
4230 \copperdefstring{italian}{monthname12} {dicembre}
4231 \copperdefstring{italian}{algorithm2eopt}{italiano}
4232 \copperdefstring{italian}{babelname}    {italian}
4233 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4234 </langit>
4235 <*numbbx>
4236 \ProvidesFile{coppe-numeric.bbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric bibliography]
4237 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4238 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4239 % the PT->EN source map and the language mappings) and only adds the [n] list
4240 % label and the numeric sorting on top.
4241 \RequireBibliographyStyle{coppe}
4242
4243 % ABNT numeric system lists references in the order they are first cited
4244 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4245 % numbered list. 'labelnumber' generates the [n] field.
4246 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4247
4248 % Bracketed labels, both in the list and for any shorthand entries.
4249 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4250 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
4251
4252 % Reference list driven by the [n] label (copied from biblalex's numeric.bbx),
4253 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4254 \defbibenvironment{bibliography}
4255 {
  \list
4256   {\printtext[labelnumberwidth]{%
4257     \printfield{labelprefix}%
4258     \printfield{labelnumber}}}
4259   {\setlength{\labelwidth}{\labelnumberwidth}%
4260    \setlength{\leftmargin}{\labelwidth}%
4261    \setlength{\labelsep}{\biblabelsep}%
4262    \addtolength{\leftmargin}{\labelsep}%
4263    \setlength{\itemsep}{\bibitemsep}%
4264    \setlength{\parsep}{\bibparsep}}%
4265    \renewcommand*{\makelabel}[1]{\hss##1}}
4266 {\endlist}
4267 {\item}

```

```

4268
4269 </numbbx>
4270 <*numcbx>
4271 \ProvidesFile{coppe-numeric.cbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric citations]
4272 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4273 % numbers [1], compressed and sorted within a single cite group. This is the
4274 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4275 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
4276 % \citet -> Author [1].
4277 \RequireCitationStyle{numeric-comp}
4278 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents — four kinds of work by three languages — each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` — CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 `example_pdfa.tex` -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo `docstrip` sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 `covers_5languages.tex` -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the `coppe` class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 `latexmkrc` -- latexmk/Overleaf build configuration

`latexmkrc` – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPE_{UF} users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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\thesubsection ... 1198	3680, 3689,	\xspace 869
	3693, 3694,	

Change History

v0.0		v3.1	
General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>symb</code>	55
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>symb</code> and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	54
<code>\maketitle</code> : Added number of examiners test.	64	v3.5	
Generalization.	64	General: New command <code>annex</code> . .	101
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english	101
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	72	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral</code> / <code>number</code> replace LaTeX's <code>roman</code> / <code>arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	60	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppforeignstring</code>) replaces scattered <code>iflanguage</code> and <code>if</code> english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	73	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETEX release. .	1		
v2.0			
General: COPPETEX release 2.0. . .	1		
v2.1			
General: COPPETEX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	57		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	54		
New approval page layout. . . .	67		
<code>\department</code> : Added new course on Nanotechnology.	55		
<code>\examiner</code> : Examiners expansion without degree.	57		

spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1	doublespacing, no folio is printed on a pre-textual sheet when something does overflow, the approval sheet holds a board of eight, and es-nosectiondot removes the period babel-spanish put in the sumario indicative.	1
New user API: titlein, usecoppelanguage, brazilianabstract environment. .	1		
\coppetexfinalpage: Initial coppetexfinalpage colophon page.	61	New options: pdfa (PDF/A-2b via pdfx, XMP built from the document), assinaturas, coorientador, rascunhoficha, listasnosumario, morewrites/semmorewrites. New command coadvisor. Under pdfTeX the class loads morewrites by itself when installed, because a work that asks for every list needs seventeen of TeX's sixteen output streams.	1
v4.1 General: Conformance with the revised 9th edition (2026) of the UFRJ/SiBI manual: continuous pagination from the folha de rosto, folio in 10pt at 2cm from the top and right edges, one-sided layout with a 3cm left margin, additional sheet with the catalog card and the CAPES collection fields, approval sheet of 3.1.2.1.3, keywords closing the three abstracts, pre-textual lists out of the sumario, Apendice and Anexo headings centred, fifth heading level, and Latin Modern instead of the bitmap fonts.	1	Single source: pdflatex coppe.ins generates everything distributed – class, biblatex styles, language packs, .bib bases, .ist, the five per-language examples, example_pdfa, the cover montage and latexmkrc. The regression suite, the adversarial documents and the harness prove the class and are NOT distributed.	1
Fixes found by the adversarial review: cover and folha de rosto compose in single spacing so they do not overflow under			